

Neck pain

by GWENDOLEN JULL and DEBORAH FALLA

It makes your arm very hot, heavy and difficult to move. Stingers don't tend to last too long: you get pins and needles down your arm as the heaviness goes away. Sometimes it's 10 seconds, sometimes maybe a minute.

England Rugby star Jonny Wilkinson, who required neck surgery after recurrent injuries

This chapter focuses on the assessment and management of non-catastrophic painful neck conditions in the athlete. While there is much attention on prevention and acute management of catastrophic neck injuries, these injuries occur predominantly, but not exclusively, in contact sports.^{1,2} Yet, many athletes suffer from debilitating non-traumatic neck pain that can affect both their participation in sport and their quality of life. Such conditions range across a spectrum from acute muscle injury to chronic repetitive strains to cervical degenerative joint disease.³⁻⁵

ANATOMICAL CONSIDERATIONS

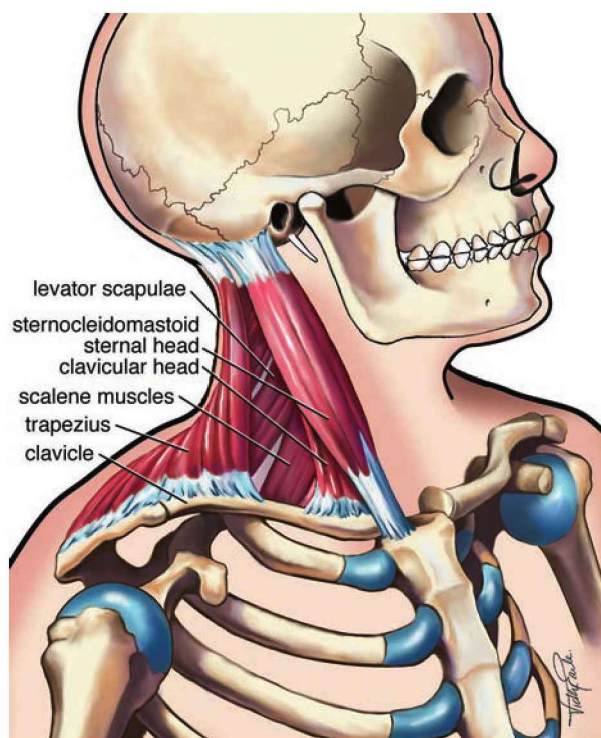
It is usually not possible to make a definitive patho-anatomical diagnosis in most neck pain conditions. Lesions can be identified on imaging with injuries such as cervical fractures or dislocations, or disc disruption (linked with symptoms of a radiculopathy), as can anatomical abnormalities which may predispose the athlete to neck pain or injury.⁶ However, for the majority of neck pain presentations in both the athletic and the general population, radiological imaging is unable to identify the pathoanatomical pain source with any certainty. As a result, the majority of disorders are generically classified as non-specific or mechanical neck pain. Despite having one label, the presentation of neck pain conditions is highly variable. Thus, the clinical examination assumes primary importance in identifying the symptom sources and any sensory, neuromuscular and sensorimotor abnormalities associated with the individual athlete's neck pain. The clinical examination relies on a foundation of anatomical knowledge.

The cervical spine is designed for movement, in contrast to the lumbar spine that is designed to carry load. As the most mobile region of the spine, its movements ensure a wide visual field. The functional cervical spine extends from the atlanto-occipital (C0-1) segment to the upper thoracic region (approximately T3-4 segment).

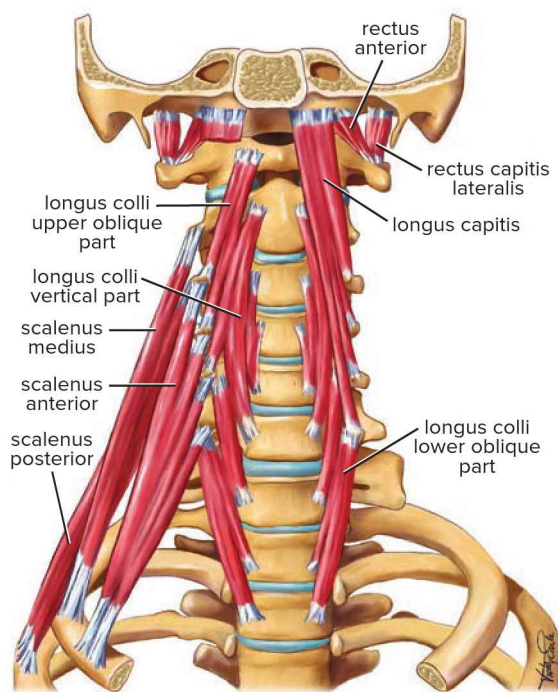
There are many key features to note when performing a clinical evaluation. Movement occurs independently and interdependently between the cranio-cervical, cervical and cervicothoracic regions. The atlanto-axial (C1-2) segment contributes about half the total rotation of the cervical region and this head movement can be performed relatively independently. Most of the remaining rotation occurs at the cervical segments (C2-7), but the head will not rotate to its full excursion unless the upper thoracic region provides the necessary 10° of this movement.⁷ From a postural perspective, there is interdependence of the lumbar, thoracic and cervical spinal curves.⁸ Evaluation must be both local and regional.

The neck muscles have many diverse roles given the broad functional requirements of the cervical region. For example, an ice hockey player needs fast movements to locate an opponent or puck, a target shooter requires fine movement adjustments for clear vision and head alignment, and a soccer player requires good stability of the neck to head a ball. To perform these tasks there must be fine muscle coordination for movement and adequate kinaesthetic awareness, as well as muscle strength.

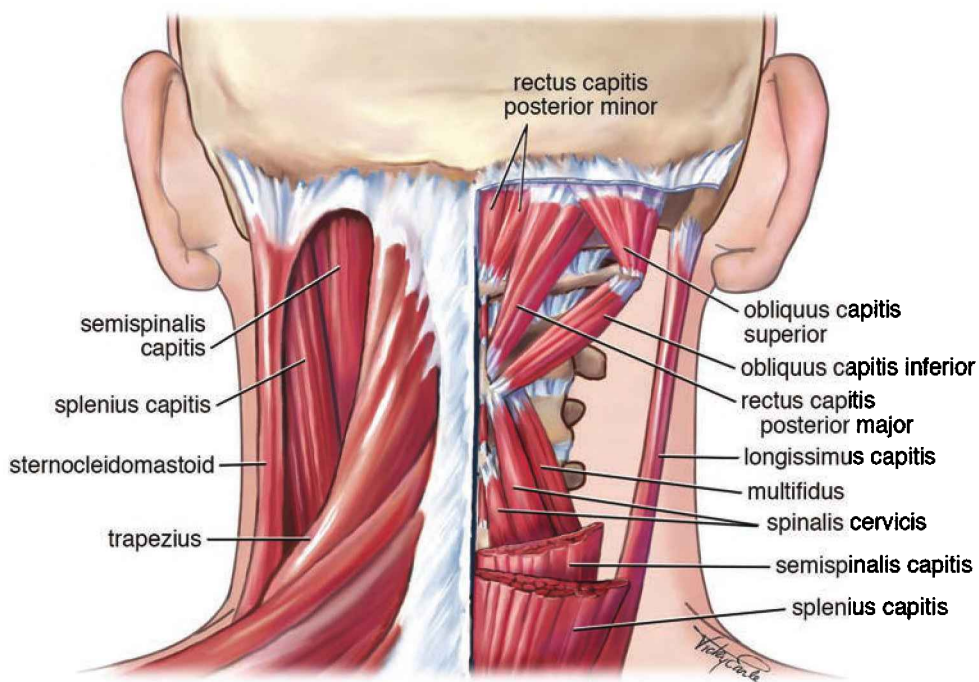
There are 44 muscles in the neck, topographically arranged to provide these functional requirements (Fig. 23.1). Some act exclusively over the cranio-cervical



(a)

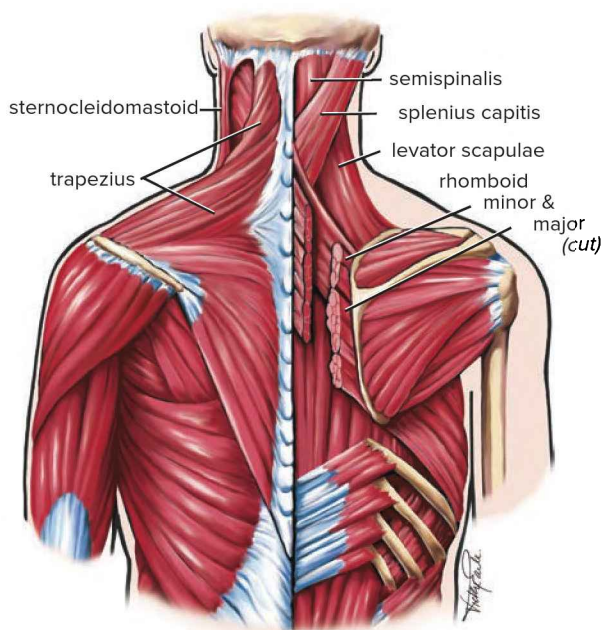


(b)



(c)

Figure 23.1 Anatomy of the neck (a) Anatomy of the anterior neck—superficial musculature (b) Anatomy of the anterior neck—deep musculature (c) Anatomy of the posterior neck—deep and superficial musculature



(d)

Figure 23.1 (cont.) (d) Anatomy of the posterior neck and scapulothoracic region

region (e.g. suboccipital muscles), others act mainly on the cervical region (e.g. semispinalis cervicis), and still others span the cranio-cervical and cervical regions with attachments from the upper thoracic region (e.g. splenius capitis). The mobile structural design of the cervical spine also relies on the vital role of the muscle system to provide stability.⁹ For stability, there is some functional specificity between the layers of both the neck flexor and the extensor muscles.

Broadly, the superficial muscles have larger lever arms and cross-sectional areas with a greater capacity to exert torque than the deep muscles.¹⁰ In contrast, the deeper muscles are more segmentally arranged with segment-to-segment attachments, larger spindle densities and muscle fibre compositions that enable them to guide and support the cervical motion segments.¹¹

Coordination between the deep and superficial muscles is required for normal neck function as the deep muscles cannot support the load of the head and the superficial muscles are not placed to support the cervical segments.¹² The cervical region also provides a point of attachment for muscles of the shoulder girdle (Fig. 23.1d) and must bear the load and absorb the forces induced on the cervical segments during upper limb function.^{13, 14}

All structures of the cervical region are innervated and thus any could be a source of nociception. These

may include the zygapophysial joints, discs, ligaments, muscles and nerves. Pain may be perceived in the neck, upper thoracic region, upper limb or head (cervicogenic headache). Various studies have mapped pain areas when discs or zygapophysial joints have been stimulated.^{15, 16}

The overlap between pain patterns of both the zygapophysial joints and discs challenges the ability to make a structural diagnosis from pain distribution alone. Furthermore, the pain distribution from symptomatic joints is more extensive than that of experimentally stimulated asymptomatic joints.¹⁷ Importantly, the suboccipital and deep cervical muscles in the neck region (multifidus and longus colli) have the highest density of muscle spindles of all muscles in the body.^{11, 18, 19} There are reflex connections between the neck and the vestibular and ocular systems (for review, see Røijezon and Treleaven).²⁰ Thus, when cervical afferentation is altered, these interconnections can underlie symptoms of unsteadiness, light-headedness or visual disturbances when associated with neck pain.

CLINICAL PERSPECTIVE

Neck pain is considered within a biopsychosocial framework that embraces biological, psychological (e.g. anxiety, fear) and social (e.g. sports or training requirements) features and their interactions that may contribute to the athlete's disorder and recovery (Fig. 23.2). Neck pain can be a recurrent condition with repeated episodes over a lifetime, with variable degrees of recovery between bouts.^{21, 22}

The Global Burden of Disease 2010 study ranked the burden of neck pain as the 4th highest in terms of years lived with disability and ranked it 21st of 291 conditions in terms of overall burden.²³ The lifetime impact of neck conditions should not be underestimated in planning rehabilitation programs for athletes. The challenge is not only to resolve an acute episode of pain to meet an immediate goal of return to sport, but to also prevent or limit recurrence for future quality of life.

Effective management of individuals with neck pain is based on good communication, a thorough history and physical examination, and sound clinical reasoning in a patient-centred approach (Fig. 23.3). The clinician carefully elicits information from the athlete and interprets it on several levels for the purposes of: ensuring patient safety (red flag conditions); making a provisional pathoanatomical diagnosis; making a definitive physical diagnosis in terms of pain mechanisms, movement, neuromuscular and sensorimotor impairments to inform a treatment plan; identifying social (sport) factors which might be driving or causing the disorder; identifying psychological features (e.g. fear of movement, anxiety)

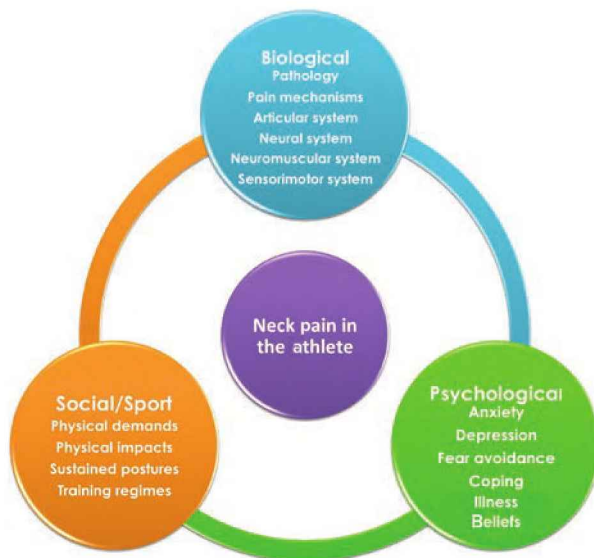


Figure 23.2 The biopsychosocial model for neck pain in sport. The biological, psychological and social components are artificially equally weighted in this diagram. The relationship is fluid. Weighting of biological, psychological and social features differ within and among domains, and within and between individuals. They also change at various time points within the course of an athlete's recovery

that might moderate recovery; and identifying possible prognostic indicators.^{24, 25} There is more certainty in decision making when the assessment reveals a clinical pattern that aligns with current knowledge of painful cervical conditions. This pattern should reflect the relationships within and between biological, psychological and social features.

History

There are four key areas of information to elucidate when taking an athlete's history. First, establish the athlete's sport and understand its requirements and implications for loads and strains on the neck region. Second, ascertain whether this is an initial onset or a recurrent neck pain condition. Third, determine the duration of neck symptoms in this current presentation (i.e. whether acute, subacute or longstanding in nature). Fourth, clarify how the neck condition began: whether symptoms arose from trauma, either direct or indirect forces (and their direction), or was insidious; and whether the onset was rapid or gradual in nature.

Identifying red flags is a vital aspect in managing athletes with cervical conditions, and the history often raises first suspicions. Except for catastrophic trauma, there are fortunately relatively few serious or life-threatening conditions that present as acute neck pain. Nevertheless, there are two conditions in particular that require clinicians' vigilance in younger to middle-aged athletes: cervical arterial dissection (CAD) and cranio-cervical ligament injury (Table 23.1). CAD can initially be misdiagnosed as a cervical musculoskeletal disorder, and the history and presentation are central to recognition of the condition. Immediate referral for medical investigation and management is essential if suspicions are raised.^{26, 27} It is unknown how frequently the alar or transverse ligaments are non-catastrophically injured. Magnetic resonance imaging (MRI) studies have found no convincing evidence of alar or transverse ligament involvement in the common motor vehicle collision and resultant 'whiplash' injury.^{28, 29} Immediate referral for medical investigation and management is essential if suspicions are raised.

In more benign cervical conditions, a detailed history can provide valuable information on adverse repetitive

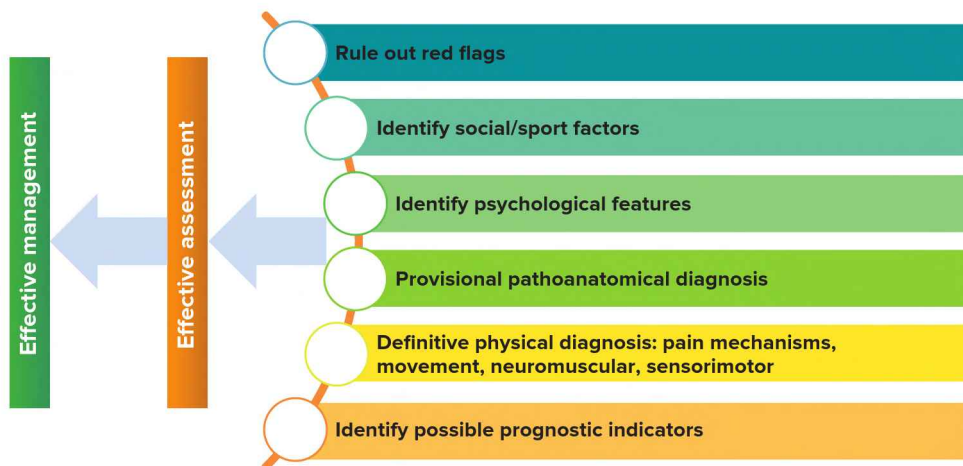


Figure 23.3 Decisions made from effective assessments will inform safe, individualised and effective management plans

Table 23.1 Red flags in acute neck pain presentations

Condition	Presentation
Cervical arterial dissection	• Acute onset 'unusual' neck pain or headache, increasing in severity over hours or days • Neurological or ischaemic signs • Recent infection or minor trauma
Cranio-cervical ligament injury	• Head or neck trauma • Acute neck pain with perioral paraesthesia, tinnitus, 'lump in the throat' or nausea

movements or prolonged abnormal postures relating to the athlete's sport or training which may be underlying the condition. If the neck pain is recurrent, determine the original time and nature of the onset and the pattern of recurrence, and track the progression of the condition.

Symptoms and their behaviour

Clinicians should obtain information about the area, nature and intensity of pain. The area of pain will give some idea of the likely segmental level that is the source of the pain. Nociception from the C0-4 segments is usually perceived in the upper cervical and possibly the head region, C4-5 in the mid to lower neck, and C5-7 in the lower neck, upper thoracic region and shoulder/arm. Identification of a particular zygapophysial or disc disorder by area of pain is not possible due to overlap of their referral patterns.¹⁵⁻¹⁷ If the athlete pinpoints the pain more posterolaterally, there is an increased possibility that a zygapophysial joint is the source of pain, as these joints are relatively superficial. Pain from irritation of nerve structures can refer into the back of the head (upper cervical nerves) or down into the shoulder and arm (C4-T2), often in a fairly typical dermatomal pattern.

The nature and intensity of the pain can give some indication of pain mechanisms. As a very basic statement, neck pain may reflect a peripheral nociceptive source (e.g. from the joints, ligaments, muscles). When there is an injury or irritation of nerves, it is termed a neuropathic pain. The former is the more common in neck pain conditions and often manifests as an ache or catching pain of mild to moderate intensity. A neuropathic pain is usually a more severe, well-delineated pain that may have lancinating and burning qualities (e.g. a 'stinger' or 'burner' caused by traction to the brachial plexus during a tackle in rugby). Pain from either source may be associated with peripheral or central nervous system sensitisation. In instances of neuropathic pain, the clinician should question the patient regarding more widespread sensitivity to touch or sensitivity to cold.^{30, 31} Pain neuroscience has advanced remarkably over the past decades. Chapters 5 and 6 provide a more comprehensive overview of pain physiology.

The presence of other symptoms should be investigated. If the athlete has a 'stinger' or a cervical radiculopathy, there may be paraesthesia or anaesthesia in the upper limb and feelings of weakness. Other symptoms such as unsteadiness, light-headedness or visual disturbances may accompany neck pain as manifestations of cervical sensorimotor disturbances. These symptoms may also be associated with concussion, a vestibular disorder or, more often in the older patient, a vascular disorder. Thus their temporal relationship with neck pain and movement should be explored. If the athlete reports headache in association with neck pain, additional information is necessary to determine whether it is cervicogenic in nature or if it is a primary headache, such as migraine or tension-type headache. Cervicogenic headaches are typically unilateral, side-consistent headaches, associated with neck pain, and pain usually begins in the neck, later developing into a headache.³² Refer to Chapter 21 for a fuller discussion on headache.

Understanding the behaviour of symptoms as well as which activities, postures or movements provoke and ease symptoms, and how they respond to rest/sleep, can confirm the mechanical nature of the condition and inform the clinician on the possible pain source. If local neck movements are the main aggravator of symptoms, the pain source may be a segmental joint dysfunction. If lifting is the primary aggravating activity, then how the axioscapular muscles are transferring the loads to the cervical segments should be examined. If the cocking phase of throwing is the main aggravator, then any link with mechanosensitivity in the upper quadrant nervous system should be investigated. The reported aggravating activities and positions should be confirmed in the physical examination, as they can direct treatment and guide re-education of appropriate postures and movements/skills pertaining to the athlete's sport or work.

Irritability of the pain and condition is also evaluated (i.e. how easily the pain is aggravated and how long it takes an acute exacerbation to settle). Sleep disturbance is also an important indicator of pain severity. High pain intensity, irritability and acuteness of the condition will limit the extent of the physical examination and the intensity of early treatment.

Psychological features

Psychological reactions to pain such as anxiety and fear of re-injury are normal responses.³³ It is important for the clinician to listen to how the athlete describes his or her pain, how he or she is dealing with it and how it has affected his or her training or sports participation to gain an impression of whether the athlete seems to be unduly anxious or is expressing any fear of movement or activity because of the neck condition. This informs the nature and extent of the assurance and education required in their management. Mild depression could be expected if the neck injury has prevented the athlete from competing in his or her sport; however, moderate or marked depression is a psychopathology requiring expert assessment and care. The clinician can also gain an impression of the athlete's pain-coping strategies (e.g. high reliance on medication or passive treatment modalities versus continuing with activities and actively coping with the pain).

Social/sport factors

It is vital for the clinician to explore and understand the requirements of the athlete's sport, training regimens and occupational activities, and their relationship to the neck pain. A formal analysis of provocative activities is mandatory (e.g. analysis of a tennis serve which aggravates/causes neck pain, analysis of the heading skill in football, analysis of cycling and bike fit, or analysis of weight-training regimens). It is also important to assess the effects

of correcting the skill on the athlete's immediate neck pain, to provide some proof of the relationship between the action and pain. This is convincing evidence for the athlete on the need to modify the skill. If the 'driver' of the pain can be removed, this increases the chances of preventing neck pain recurrences. However, the nature of certain sports makes it difficult to remove the 'driver'. For example, modifications of spinal postures and head positions may be limited in road cycling. It will be important to determine tissue tolerance to these postures and develop a program that relieves the strain on these structures before this threshold for pain is reached. This analysis of the symptom-provoking aspects of the sport is crucial for any long-term solutions for the athlete's neck pain.

Patient-reported outcome measures

There are numerous patient-reported outcome measures (PROMs) recommended for the assessment of neck pain. Three common measures are the Numeric Pain Rating Scale (NPRS),³⁴ the Neck Disability Index (NDI)³⁵ and the Patient Specific Functional Scale (PSFS).³⁶ They are described in detail in Table 23.2. PROMs allow the clinician to understand the pain and disability level of the athlete in relation to all individuals with neck pain (NPRS and NDI), as well as to understand the specific functional concerns of the individual (PSFS). PROMs are essential to evaluate the course of recovery throughout treatment.

Table 23.2 Patient-reported outcome measures (PROMs) in neck pain

Measure	Description	Minimum clinically important difference (MCID) ³⁷	Minimal detectable change (MDC) ³⁸
Numeric Pain Rating Scale	Quantifies self-reported pain levels and measures pain changes over time. Athletes rate their pain on a scale from 0 and 10; 0='no pain'; 10='the worst pain imaginable'	1.3 points ³⁹	3 points ⁴⁰
Neck Disability Index	Ten items on functional abilities and symptoms. Each item has a 5-point scale and the score is summed and converted to a percentage	9.5 points (19%) ³⁹	10.2 points (20%) ⁴¹
Patient Specific Functional Scale	Athletes nominate at least 3 activities where they are experiencing difficulty as a result of their condition. For each activity, they rate their ability to perform it, with 0='completely unable' and 10='able to complete the task at full, pre-injury level'	2.0 points ⁴¹	2.1 points ⁴¹

MCID: The smallest difference that patients perceive as beneficial

MDC: The amount of change that must be observed before the change can be considered to exceed the measurement error

Imaging

When the neck pain has resulted from trauma (e.g. collisions, crashes, falls), imaging (plain radiography, computed tomography [CT] scans) is indicated to identify or rule out such lesions as fractures or subluxations/dislocations. There are clear protocols (Canadian C-Spine Rule; NEXUS Criteria) to guide the need for imaging after trauma which have proven sensitive, although their specificity has been challenged.^{42, 43} This can affect adherence to these guidelines, as clinicians and athletes want to ensure that serious pathology is not missed. Imaging is also indicated when, on clinical grounds, there is any suspicion of a lesion or fracture.

There is long-standing debate about the need for routine imaging of people presenting with neck pain because the majority of routine imaging provides no additional or certain information to assist structural diagnosis. Even recent studies using MRI have failed to detect and define differences between the non-pathological age-related changes of the cervical spine in people without neck pain and the cervical spines of people with neck pain conditions.^{28, 44, 45} Thus routine diagnostic imaging is not indicated for most people with neck pain. The decision to order imaging for the athlete with a neck pain condition should be based on clinical grounds to confirm a clinical diagnosis or clinical suspicion.

Physical examination

The clinical reasoning process commenced in the history continues throughout the physical examination. The physical examination further defines the condition and has three principal aims:

1. to gain a clear picture of the immediate source(s) of pain and the nature of the dysfunction (e.g. a painful cervical segment, protected by local muscle spasm that is restricting segmental and regional motion)
2. to detect and document the effect of pain and injury on the sensory, neuromuscular and sensorimotor systems, and gain an understanding of the interactions between these systems
3. to develop a management program directed by the findings of the physical examination; this management program may be moderated by knowledge of history of onset and pain intensity/mechanisms.

No single test will provide the answer; rather, the clinician seeks a pattern of test responses from the various systems. Two clinical examples illustrate this important point.

The first is the athlete who reports that neck and right arm pain is provoked in the early cocking phase of throwing. Nothing abnormal was detected in the glenohumeral complex. It was reasoned that a cause of

this pain could be mechanosensitivity of nerve tissue of the upper quadrant. The pattern of signs that would fit this hypothesis⁴⁶ is shown in the box below. If such a pattern of physical impairment and pain responses was not elicited, the clinician would need to reflect on his or her provisional diagnosis and propose an alternative hypothesis to test.

The second is an athlete who presents reporting headaches that are increasing in frequency and who believes they are caused by his or her neck (the athlete has neck pain and tenderness with the headache). The clinical confounder is that most primary frequent intermittent headaches (migraine and tension-type headache) have some neck pain and tenderness in association with headache.⁴⁷ This reflects the bi-directional pathway of the trigeminocervical nucleus.⁴⁸ A cervicogenic headache is classified as a secondary headache (i.e. secondary to a primary cervical musculoskeletal dysfunction). If the athlete's headache is cervicogenic, then it would be expected to be associated with a pattern of cervical musculoskeletal dysfunction: cervical movement impairment, segmental joint dysfunction (C0-4 which refers pain into the head) and muscle function impairment.⁴⁹ Neck tenderness as a single sign has no diagnostic significance. Failure to find this pattern of cervical musculoskeletal dysfunction would rule out a cervical cause of headache. This chapter does not cover the temporomandibular joint complex. Nevertheless, the clinician is reminded of the role that craniomandibular dysfunction may play in head, face and neck pain.

There are many tests that may be employed in the physical examination to evaluate the function and integrity of articular, neural, neuromuscular and sensorimotor systems of the cervical region. The tests used will depend on the individual athlete. For example, there is no indication to use the upper limb neurodynamic tests or Spurling's test in an athlete presenting with right upper cervical pain aggravated by looking upwards. Likewise, in acute injury or pain, examination will be curtailed to those tests absolutely necessary to gauge severity and to inform initial management. The order in which tests are conducted in the physical examination usually considers the athlete's position and comfort. However, for ease of description here, tests will be described in groups rather than the ideal order of application.

The physical examination involves the following.

1. Postural analysis
 - a. dynamic postures (symptom provoking positions or relevant sports/activity-related positions)
 - b. static posture of the spine and axioscapular region in sitting and standing in all three planes
 - c. axioscapular posture.

An example of clinical reasoning in the physical examination to confirm the hypothesis that an athlete's neck and arm pain while throwing was related to mechanosensitivity of nerve tissue of the upper quadrant

Functional complaint: Neck and (R) arm pain provoked in the early cocking phase of throwing

Shoulder joint: No abnormality detected in any test

Reasoning: Pain from mechanosensitivity of nerve tissue of the upper quadrant

Physical examination confirmation

Posture: Slightly elevated (R) shoulder

Scapular depression reproduced slight arm pain (stress on brachial plexus)

Active/passive movement dysfunction: Cervical lateral flexion (LF) to the left most provocative of arm pain (strain on right nerves)

Range of lateral flexion is restricted (compared to normal) and right arm pain is present when the nerves are pre-tensioned

Arm pain ↓ when arm passively supported in 90° abduction and external rotation with elbow flexed (nerves off tension)

Adverse responses to neurodynamic tests:

Upper limb neurodynamic test

– lacks 30° elbow extension and neck and arm pain reproduced

– cervical LF (L) increases pain; LF (R) decreases arm pain

Evidence of a local area of pathology: Manual examination: painful segmental joint dysfunction C5–6

2. Active movement tests—each assessed for cranio-cervical (C0–2), cervical (C2–7) and cervicothoracic (C7–T4) regions
 - a. flexion
 - b. extension (Fig. 23.4)
 - c. lateral flexion
 - d. rotation
 - e. flexion/rotation (Fig. 23.5).
3. Quantitative sensory testing (e.g. pressure pain threshold, thermal pain threshold)
4. Manual examination of segmental dysfunction (e.g. posterior-anterior glides [Fig. 23.6], PPIVMs)
5. Tests of the nervous system
 - a. clinical neurological examination (if indicated)
 - b. upper limb neurodynamic tests (movement and mechanosensitivity of nerves)—Chapter 15.
6. Tests of muscle function
 - a. cervical flexors
 - i. cranio-cervical flexion test (Fig. 23.7)
 - ii. cervical flexor strength and endurance.
 - b. cervical extensors
 - i. cranio-cervical extension test (Fig. 23.8a)
 - ii. cranio-cervical rotation test (Fig. 23.8b)
 - iii. cervical extensors (biased towards the deep cervical extensors) (Fig. 23.8c)
 - iv. modified Biering-Sorensen test.
 - c. axioscapular
 - i. dynamic scapular control (Fig. 23.9)
 - ii. lower trapezius, serratus anterior in prone and quadruped
 - iii. muscle length tests (e.g. levator scapulae, pectoralis major/minor).
7. Tests of cervical sensorimotor function
 - a. test of joint position sense (Fig. 23.10)
 - b. test of movement sense (Fig. 23.11)
 - c. static and dynamic balance
 - d. ocular motor tests (gaze stability, smooth pursuit neck torsion test [SPNTT] and evaluation of rapid eye movements).
8. Additional tests (if indicated)
 - a. cervical arterial dysfunction
 - b. additional passive movement tests when indicated (e.g. Spurling's)
 - c. cranio-cervical ligament tests (e.g. Sharp-Purser test; alar ligament tests).

Postural analysis

One of the most important assessments is the analysis of the relevant sports/work or activity-related postures and movements that aggravate the athlete's pain. The athlete demonstrates the activity or position that is pain provoking. The clinician analyses the movement and its control from local and regional perspectives. The clinician should facilitate correction of the posture, movement or technique to gauge the effect on pain or performance. This will accurately and relevantly inform the management approach.

Posture is then formally examined in standing and sitting. The regions of the spine are interdependent, and assessment of head and neck orientation can only be made in relation to orientation of the lumbopelvic region and thoracic spine. The most common poor sitting posture is the slumped 'flexed' posture which facilitates a forward head posture. Some persons have an extended spinal

posture in sitting (e.g. some ballet dances and gymnasts) while others have a good 'neutral' posture. It is necessary to determine whether the person's 'non-ideal' posture is contributing to their neck pain.

An answer can be gained by a simple test-retest approach. Test cervical rotation in the person's natural posture and note any movement restriction or reproduction of their neck pain. Correct posture to the ideal upright neutral position and retest cervical rotation. If the range and pain response improve significantly, this suggests that poor posture is one driver of the neck pain or loss of function and indicates that postural re-education will be an important strategy in management.

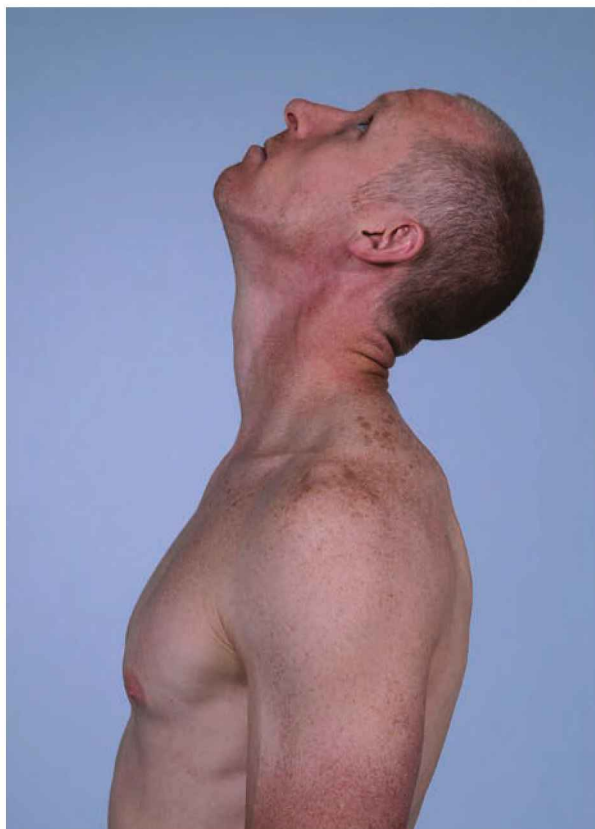
At the other extreme, if there is no change in rotation range or pain with postural correction, then the clinician can reason that posture is not a major contributor. In this case, postural correction will not be taught with a rationale that it is a driver of pain. As will be discussed in the management section, postural correction will be taught, but with the intent of facilitating the postural supporting muscles.

A similar process is undertaken in the evaluation of scapular orientation and this is described more fully in the later section on axioscapular muscles.

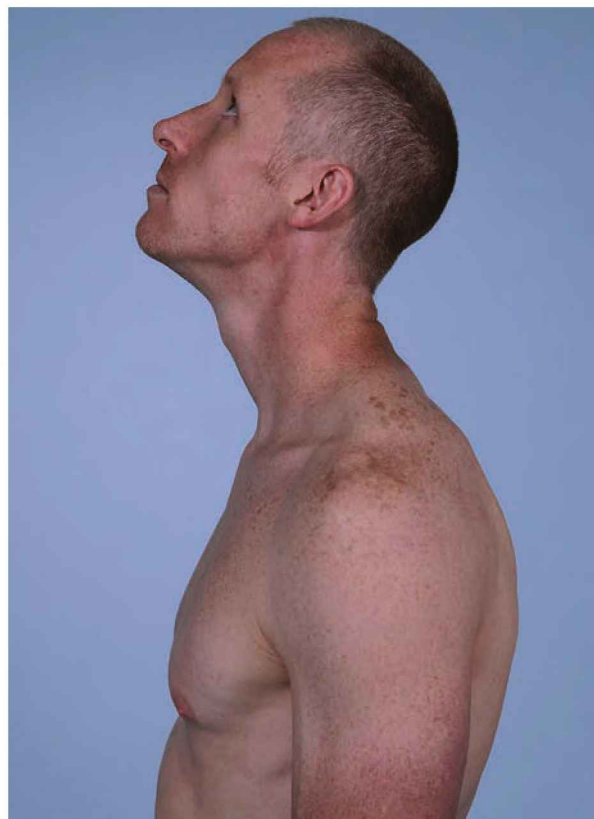
Active movement tests

The observation of reduced cervical range of movement is valuable in the diagnosis of cervicogenic headache, cervical radiculopathy and cervical spine injuries.⁵⁰ Movement is an impairment in most neck pain conditions. Movements of flexion, extension, lateral flexion and rotation should be assessed in the three functional regions of the cervical spine: cervical (C2-7), cranio-cervical (C0-2) and cervicothoracic (C7-T4) regions. The information gained from this movement analysis includes range of motion, pain with motion and the ability to control motion. Movements can be affected by articular, muscular or neural tissue, and differentiation of the source of restriction or pain will inform treatment planning.

In the examination of cervical flexion, ensure that the chin stays in a neutral position since anatomically, full



(a)



(b)

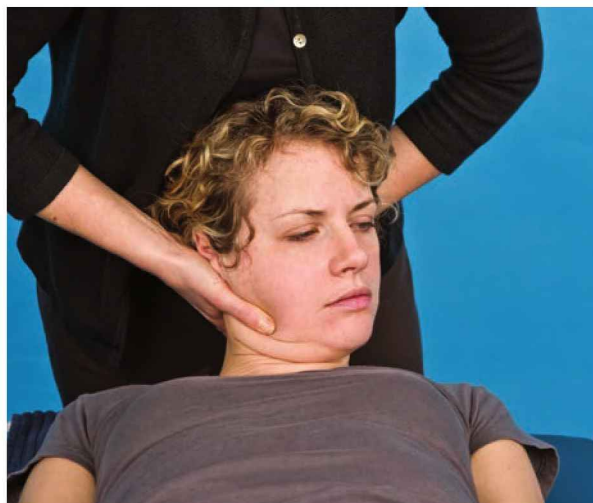
Figure 23.4 (a) Cervical extension. Note the displacement of the head behind the line of the shoulders (b) Cervical extension. If movement is only occurring in the upper cervical region, there is no posterior displacement of the mid to lower cervical segments

cervical flexion cannot be performed with simultaneous full upper cervical flexion. Extension is informative in detecting both articular and motor control problems. Normally, the head should displace behind the line of the shoulders in full extension. While the person's face may be directed to the ceiling (upper cervical extension), it is often observed that the mass of the head is not displaced posteriorly to a sufficient extent (Fig. 23.4). This may be a protective strategy for a painful cervical joint, or masking poor muscle control of the region (or both). Encouragement is given to perform the movement correctly and the response noted. If pain limits the movement, the movement usually stops at the segmental source of the pain. If the problem is poor control (i.e. poor eccentric control of the deep cervical flexors), a 'give' in the mid-cervical region may be observed as the neck extends, or the patient may protrude their chin as the initiating action on return from extension to the upright position (due to poor concentric control of the deep cervical flexors).

Assessment of axial rotation gives a clear indication of the contribution of the cranio-cervical, cervical and cervicothoracic regions to total head rotation. Cranio-cervical rotation can be tested in relative isolation from the remainder of the cervical region with the flexion-rotation test (Fig. 23.5a).^{51, 52} The contribution of the cervicothoracic region⁷ to total rotation can be assessed through observation, and the segmental contribution to the movement can be determined by palpation of the movement at each spinous process as the patient repeatedly rotates his or her head (Fig. 23.5b). The clinician can detect segments that are hypomobile and not contributing to the movement.

Lateral flexion is probably the most useful movement to differentiate between an articular, neural or muscular restriction to movement (see box above).

Despite difficulties in imaging relevant pathoanatomy, selected movement tests can identify facet joint dysfunction. The flexion-rotation test has been proven to be a reliable test with content and discriminant validity.⁵¹⁻⁵⁵ It can be used with confidence to locate C1-2 joint dysfunction, especially in the diagnosis of cervicogenic headache. The extension-rotation test,⁵⁶ when used in combination with a palpation and manual



(a)



(b)

Figure 23.5 (a) The cervical flexion rotation test (C1-2 rotation). The athlete is positioned in supine and the head is supported by the clinician. The athlete is asked to look down at his/her toes (cervical flexion) and simultaneously the clinician moves the athlete's neck into full flexion. The head is then rotated to the left and right and the clinician is assured of performance of a correct test when there is a definite end to the movement. There should be at least 30° of rotation, and a side-to-side difference of 10° is considered a positive test result⁵³ (b) Assessment of cervicothoracic rotation. The athlete actively and repeatedly rotates his/her head. Using a gentle lateral glide on the spinous process in time with the active movement, the clinician perceives the presence (or not) of rotation of each thoracic vertebra to approximately the T4 level

examination, is able to detect the painful zygapophysial joint—as tested against facet anaesthetic blocks.

Quantitative sensory testing

Many neck pain conditions involve peripheral nociception with or without peripheral sensitisation. Of particular concern are athletes who present with central nervous system sensitisation. Care must be taken in both the examination and management approach for these individuals as overzealous examination and management may easily aggravate their condition. Examination and management techniques should cause no further pain. Conditions in which central sensitisation may occur include cervical radiculopathy,³⁰ cervicogenic headache⁵⁷ and whiplash-associated disorders (WAD).⁵⁸ Such individuals may present with many sensory disturbances including widespread mechanical hyperalgesia and cold hyperalgesia. Pressure pain thresholds may be tested with an algometer or the clinician could determine if the athlete seems oversensitive to touch, not only over the neck but also more distally in the arms and legs. A simple clinical test can test cold hyperalgesia.⁵⁹ If patients report pain >5/10 on a NPRS after a 10-second application of an ice pack to the neck region, the clinician could be suspicious of the presence of cold hyperalgesia.

Passive tests

Manual examination of segmental dysfunction

Manual examination is used to detect the painful segment(s), and to qualitatively assess segmental motion and tissue resistance to the manually induced movement ('end feel'). The segments from C0-1 to the upper thoracic region are examined. Posterior-anterior glides (Fig. 23.6) can be regarded as a gentle provocative test to locate the symptomatic facet joint or cervical segment. Passive physiological intervertebral movements (PPIVMs) are used to both confirm the segment(s) with altered mobility and to understand the nature of the movement impairment. Although not without historical controversy, manual examination is proving to be an accurate method to locate the symptomatic segment(s). Manual examination with and without other physical tests accurately identifies segmental levels of dysfunction and assists in differential diagnosis of disorders such as cervicogenic headache.^{49, 56, 60}

Other passive movement tests (when indicated)

There are several additional testing procedures that can be employed when necessary in the examination of an athlete with a neck pain condition. Some clinicians extend the examination to direct a particular treatment approach; for example, examining movements in combination or gauging the effect of repeated movements.^{61, 62} Other



Figure 23.6 Posteroanterior glides on the C2–3

zygapophysial joint. Gentle manual forces are applied over the lamina of the zygapophysial joints of all segments as well as over the spinous processes to determine which is the painful segment associated with the athlete's neck pain disorder. Good technique is required and the posteroanterior glides must be applied painlessly to avoid false positive results

tests seek a particular pathoanatomical compromise, such as Spurling's test. This test, a combination of cervical extension and lateral flexion, narrows the intervertebral foramen and is used to determine if arm symptoms are related to a sensitive and/or compromised cervical nerve root.⁶³ There are clinical tests for the cranio-cervical ligaments, specifically the transverse ligament, tectorial membrane and alar ligament, as well as the integrity of the cervical segments. Such tests would be indicated after trauma. A variety of symptoms are proposed that might arouse suspicion, such as feeling unable to hold the head up or a desire for support either by a collar or hands.⁶⁴ The test for the transverse ligament (Sharp-Purser test) has been validated in a population with rheumatoid arthritis.⁶⁵ MRI has established the construct validity of both the side-bending and rotation clinical stress tests for alar ligament integrity.⁶⁶ However, clinicians should consider that the true clinical utility of these tests has yet to be proven and the true incidence of non-catastrophic cranio-cervical ligament injury is yet to be established.^{28, 29}

Tests of the nervous system

Clinical neurological examination

A clinical neurological examination (sensation, muscle strength and reflexes) is indicated when there is suspicion of a cervical radiculopathy and compromised nerve conduction (i.e. athlete presents with referred arm pain or has suffered a 'stinger' due to traction of the brachial plexus or compression of the cervical nerve roots).

Tests of movement and mechanosensitivity of nerve structures

Nerves and their surrounding structures are pain sensitive and can contribute to neck pain conditions. The classic upper limb neurodynamic test (see Chapter 15) is a valid and reliable test of brachial plexus mechanosensitivity.^{46, 67-69} The test is considered positive when there is at least partial reproduction of symptoms and these symptoms can be altered via structural differentiation.⁶⁷ Further, there is usually a side-to-side difference in the range of movement in the final test component (usually elbow extension) with increased muscle resistance to limit the provocative movement. As illustrated earlier (see Box earlier in the chapter), there is usually a pattern of clinical symptoms and signs that nerve tissue mechanosensitivity is a pain source.

Tests of muscle function

There are potential changes in control and structure of neck and axioscapular muscles in the presence of neck pain and injury. Changes in motor control include altered coordination between deep and superficial neck muscles, which may compromise the deep muscles' capacity to support the cervical segments in posture and movement.¹² Evidence of this impairment is strong for the flexor muscles⁷⁰⁻⁷² and is emerging for the extensor muscles.^{73, 74}

Altered muscle coordination has been found in persons with neck pain of both insidious and traumatic onset,^{70-72, 75} indicating that it is common motor dysfunction (analogous to the inevitable quadriceps muscle dysfunction that occurs with knee conditions regardless of cause). Muscles lose directional specificity,⁷⁶ and increased co-activation of the neck flexor and extensor muscles during isometric and functional tasks has been found.⁷⁵⁻⁷⁷ Onsets of muscle activity are delayed, as demonstrated by an increased latency between the onset of the deltoid muscle and onset of the neck muscles with rapid arm movement,^{78, 79} and by delayed activation of the neck muscles in response to full-body perturbations.⁸⁰ This has implications for unprotected loading on cervical segments during sport and functional activities. People with neck pain also display reduced strength and endurance at various contraction intensities, and their neck muscles may demonstrate increased muscle fatigability.⁸¹⁻⁸³

Muscle biopsies reveal changes in fibre type within the neck flexors and extensors in people with longstanding neck pain.⁸⁴ Specifically, there is an increased proportion of type IIC transitional fibres, consistent with a transformation of slow-twitch oxidative type I fibres to fast-twitch glycolytic type IIB fibres, which aligns with the cervical muscles' decreased endurance capacity. Muscle atrophy has been documented in individuals with chronic neck pain conditions⁸⁵⁻⁸⁷ and widespread fatty infiltrate

has been identified in the flexor and extensor muscles, particularly in the deep muscles of those with moderate to severe chronic WAD.⁸⁸ It appears that fatty infiltrate may be a feature of chronic whiplash, but not of chronic, insidious-onset neck pain,^{89, 90} which suggests there might be differences in pathophysiological mechanisms between these two neck pain groups.

The axioscapular muscles are commonly tender to palpation but more importantly, poor muscle function of the shoulder girdle may overload cervical structures and contribute significantly to pain. The upper trapezius attaches to the cranium and ligamentum nuchae, and the levator scapulae attaches directly to the upper four cervical segments. Normal upper limb function induces both compressive loads and movement in the cervical segments.^{13, 14} Poor scapular muscle control may adversely increase these loads and strains.

Electromyographic studies have revealed greater fatigability of the upper trapezius⁹¹ and altered patterns of trapezius muscle function in functional tasks in individuals with neck pain disorders.^{75, 92} Moreover, the spatial distribution of upper trapezius muscle activity changes in the presence of pain.^{93, 94} Evidence for serratus anterior dysfunction in neck pain patients is also emerging.⁹⁵ A link between neck pain and altered axioscapular muscle function has recently been established in an experimental pain study, where the neck pain resulted in some reorganisation in axioscapular muscle activity.⁹⁶

Assessment of cervical flexors

The cranio-cervical flexion test (CCFT) assesses the activation and holding capacity of the deep longus capitis and colli muscles (for review see Jull et al.).⁹⁷ It is a low load test performed in supine. The test is conducted in two stages (Fig. 23.7).

The first stage evaluates motor control. The athlete performs five progressive stages of increasing cranio-cervical flexion range and is guided to each stage by feedback from a pressure sensor (such as the Stabilizer Pressure Biofeedback unit) placed behind the upper neck. The pressure sensor indirectly monitors the slight flattening of the cervical curve that occurs with longus colli contraction. The displacement, registered by the pressure sensor, permits some quantification of deep cervical flexor muscle performance.⁹⁸ The assessment determines the pressure level that the athlete can achieve in the test when asked to target each of the five stages of 2 mmHg pressure increments from a baseline of 20 mmHg to a maximum 30 mmHg. A 'correct' performance is when the athlete performs a slow and controlled cranio-cervical flexion action (rather than retraction) and no excessive activity of the sternocleidomastoid (SCM) or anterior



(a)

Figure 23.7 (a) The cranio-cervical flexion test (CCFT). The pressure sensor is placed suboccipitally and inflated until a stable baseline of 20 mmHg is gained. In stage 1 of the test, the athlete is instructed to feel the back of his or her head slide up the bed to nod his or her chin. The athlete repeats the action five times, progressively targeting increasing increments of 2 mmHg from 22–30 mmHg, holding each pressure target for 1–2 seconds and relaxing between each. The clinician analyses the quality of cranio-cervical flexion movement at each point. Each pressure target should be achieved with increasing ranges of flexion, which will not be observed if the patient substitutes with a retraction action. Any excessive use of the superficial flexors should be observed and palpated. In stage 2 of the test, the athlete nods to the target pressure and holds the contraction. In an acceptable performance, the athlete uses the correct cranio-cervical flexion action and can hold the pressure steady. Inadequate performance is judged when the patient cannot hold the pressure steady (the pressure drops off target), it is held with a jerky action, or dominant activity of the superficial flexors is observed

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(b)

Figure 23.7 (b) The face of the pressure sensor showing the marked 2 mmHg increments that the patient targets in the CCFT

scalene (AS) muscles is observed. Note the SCM and AS muscles flex the neck but not the head, so they should not be excessively active during cranio-cervical flexion.

Stage 2 tests the endurance of the deep cervical flexors at functionally relevant low contractile intensities. Stage 2 is performed when the patient can correctly perform the cranio-cervical flexion action; thus, this stage may be tested at the second treatment session, once the athlete has mastered the action.⁹⁷ In a formal assessment of the endurance capacity, the athlete performs 10 repetitions of 10-second holds at each progressive pressure level from 22 mmHg until failure. Most asymptomatic individuals perform well to at least 26 or 28 mmHg whereas individuals with neck pain can usually only achieve stages 22 or 24 mmHg.⁹⁷ As a clinical note, the formal assessment of endurance can take up to 5 minutes or more to perform, which is a long time in a busy practice. The clinician can take a shortcut and use the test to determine at which pressure level the athlete can commence training. In this case, the clinician judges the athlete's performance on approximately three repetitions of 5-second contractions at 22 mmHg. If performance is satisfactory, the test is quickly progressed to 24 mmHg and so forth, until the patient shows difficulty in maintaining a steady contraction. Training commences at the level below the stage of failure.

Strength and endurance of the neck flexors (both superficial and deep flexors) are tested at a time only when the patient's pain and muscle performance level permits. These are tested with a head lift task while maintaining a cranio-cervical flexion position. This is a test predominantly of the SCM and AS muscles, which provide more than 80% of the cervical flexion torque.^{10, 99}

If the patient's chin protrudes when attempting to lift or hold the head position, this means the deep cervical flexors are too weak to cope with the level of load.¹⁰⁰

Assessment of cervical extensors

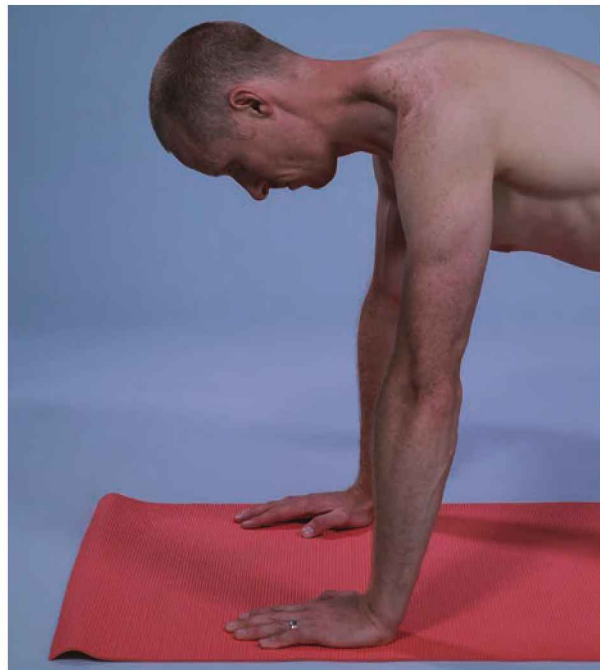
Impairment in the suboccipital muscles is common and recent research is suggesting that the deeper cervical extensors may have greater impairments than the superficial extensors.^{73, 74} Hence, these muscles are of particular interest in clinical tests. The topographical arrangement of the extensor muscles allows some targeting of these muscle groups.

The athlete is positioned in four-point kneeling or lies prone, propped on elbows, with the cranio-cervical region in a neutral position.¹⁰¹ All extensors work in this position to support head mass, but the suboccipital muscles are targeted with a test of cranio-cervical extension (chin lift and lower, rectus capitis posterior major and minor) and cranio-cervical (C1-2) rotation (rotate 30-40°, obliquus capitis superior and inferior) (Fig. 23.8a & b). Often no obvious impairment is observed due to the low load nature of the test. However, in athletes with an upper cervical disorder, it is not uncommon to observe that the athlete has difficulty in actually performing rotation at C1-2, and rotates through the mid and lower cervical region. This deficit should be remediated in the exercise program.

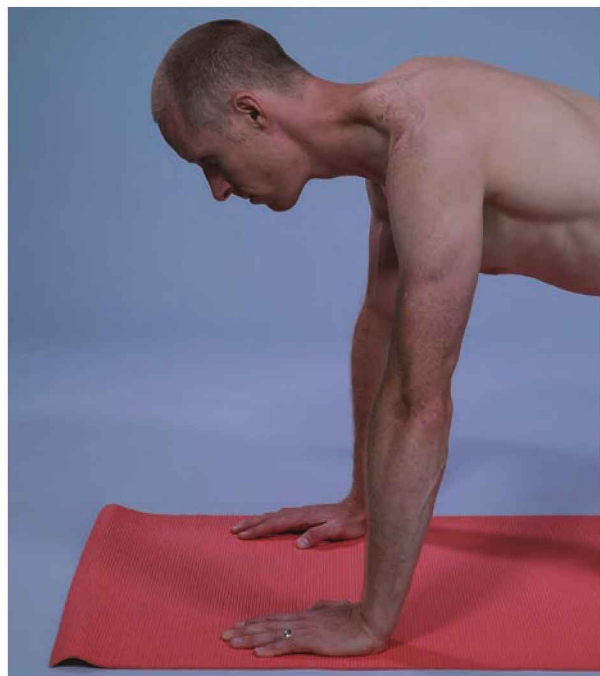
The deeper cervical extensors, semispinalis cervicis and multifidus, are targeted by testing cervical extension with the cranio-cervical region maintained in a neutral position.^{102, 103} The superficial extensors attach to the cranium and the neutral head position takes away some of their mechanical advantage. The axis of motion for cervical extension passes through the lower cervical region.

To test cervical extension of the deeper muscles, the athlete flexes his or her head and neck to look at the knees and then curls the head and neck back into extension (Fig. 23.8c). The neutral cranio-cervical region is maintained with eye facilitation. The athlete pretends that there is a book between his or her hands, keeping his or her eyes on the book while curling the head into extension. Neck extension should be around 30°. An inadequate performance occurs when the athlete can only extend to the neutral position, reports fatigue after a few repetitions, or cannot extend without use of cranio-cervical extension (i.e. the superficial extensors).

The second phase of strength and endurance testing for the neck extensors is performed once the athlete can perform multiple repetitions of cervical extension in the four-point kneeling position, and only as their pain permits. The modified Biering-Sorensen test can be used.¹⁰⁴ The athlete lies prone with his or her head supported over the end of the couch such that the cranio-cervical and cervical regions are in a neutral position.



(a-i)



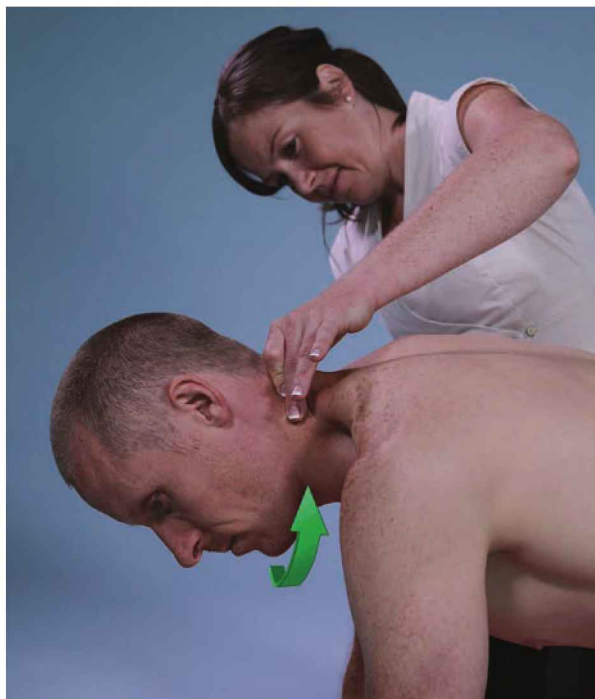
(a-ii)

Figure 23.8 (a) The suboccipital extensors are tested with the movement of head extension (a 'yes' nodding action). The cervical spine should remain in neutral and the athlete should be able to perform discrete cranio-cervical extension. The neutral starting position is shown in (i), the cranio-cervical extension position in (ii).

Support is released and the time for which the athlete can hold the head/neck position without deviation is recorded. The test can be made more time-efficient if a light weight (e.g. 1 kg) is placed on the head and baseline values recorded. The test can be extended to evaluate the thoracic extensors by having the athlete raising their shoulders from the supporting surface.

Axioscapular muscles

It is the interaction of several muscles that control the movement of the scapulae. Clinical assessment of scapular muscle function involves analysis of scapular posture at



(b)

Figure 23.8 (cont.) (b) The suboccipital rotators are tested with the movement of head rotation (a 'no' action). The cervical spine should remain in neutral. Movement should occur as a head rotation only, without rotation in the mid lower cervical spine. The clinician may gently fix C2 to help the patient localise the C1–2 rotation (c) The cervical extensors are tested by performing cervical extension while maintaining the cranio-cervical region in a neutral position. This neutral position is attained and maintained through eye facilitation. The athlete first curls his/her head down to look towards the knees (i). As the athlete curls the head and neck in extension, the eyes are kept fixed on an imaginary book located between the hands, which will encourage the cranio-cervical neutral position (ii). Good cervical extension is demonstrated here, but many athletes cannot extend past the neutral position



(c-i)



(c-ii)

rest, under light load with arm movement, as well as by using formal tests of the muscles' contractile abilities and extensibility.

Variation in scapular orientation is common and examination is necessary to disentangle normal variations from scapular dysfunction contributing to the athlete's neck pain. There are several common scapular positional faults.¹⁰⁵

A downwardly rotated scapula ('ski slope' appearance of line of the nape of the neck) may be due to poor function of the three portions of the trapezius and overactivity or tightness of the levator scapulae muscle. Medial border winging of the scapula may be caused by weakness of serratus anterior muscle, and prominence of the inferior scapular border because of excessive anterior scapular tilt may be due to weakness of the lower trapezius muscle. There may be increased resting tone and tenderness to palpation of the levator scapulae and rhomboid muscles if they are compensating for poor upper trapezius function.

An indication of a direct relationship between scapular posture, poor axioscapular muscle function and the neck pain can be assessed with a simple test (Fig 23.9).^{101, 106} First, the athlete's baseline cervical range of motion and pain is assessed. Next, the clinician manually corrects scapular position and the athlete repeats the cervical rotation movement. An improvement in range and/or pain signals that poor scapular posture and muscle dysfunction is likely contributing to the cervical pain or restricted movement and warrants attention in treatment.

There is minimal scapular motion within the first 30–40° of arm elevation.¹⁰⁷ An initial impression of dynamic scapular control in this neutral posture can be



Figure 23.9 Testing the effect of manually correcting the scapula to a neutral position on range and pain in cervical rotation. Improved range and reduced pain indicates a role for aberrant scapular posture in the neck pain condition

gained by applying very light isometric resistance to shoulder flexion, abduction and external rotation while observing any subtle movement of the scapula. With resisted flexion, an inability to maintain posterior tilt of the scapula would suggest weakness in the lower trapezius and serratus anterior muscles. With resisted abduction, an inability to maintain an upward rotation of the scapula (i.e. there is a slight depression of the acromion) suggests weakness of the upper trapezius muscle. With resisted external rotation, an inability to maintain external rotation of the scapula suggests weakness in the serratus anterior and lower trapezius muscle. In summary, with assessment of arm elevation, impaired trapezius and serratus anterior muscle function is revealed by a reduced ability to posteriorly tilt, upwardly rotate and externally rotate the scapula during full arm elevation as well as by poor eccentric control with arm lowering.

Control and endurance of the axioscapular muscles relevant to their postural function is tested in prone lying under low load, using a modified grade 3 muscle test for the lower trapezius¹⁰⁸ (i.e. the arms rest on the bed by the side with the elbow in 90° flexion). One side is tested at a time. The athlete's scapula is passively placed in a neutral position on the thorax and the athlete holds the position. Muscle activity to hold the scapula is analysed by observation and palpation. Signs of compensation for poor axioscapular muscle activation are noted such as adduction and extension of the arm via compensation by the latissimus dorsi. Signs of fatigue (tremor or loss of position) are also noted. The capacity of the axioscapular muscles to fix the scapula to the thorax in a neutral position is also tested under load in the quadruped position. Substantial winging of the medial border of the scapula is an indication of poor axioscapular muscle control, particularly of the serratus anterior muscle.

A protracted scapula may indicate that length tests of levator scapulae or pectoralis major and minor muscles are warranted. Muscle length tests should not be performed when the neck pain is acute or in the presence of mechanosensitive neural tissue. Testing may aggravate the neck condition in these circumstances, as the upper trapezius and the scalene muscles provide protection to sensitive nerve structures. The upper trapezius muscle is a common sight of pain and tender points; however, the muscle is often in a lengthened position (in association with a downwardly rotated scapula). Stretching is not indicated in these circumstances; rather, rehabilitation should target its contractile capacity, along with the other portions of trapezius and serratus anterior muscle.

Tests of cervical sensorimotor function

The cervical sensorimotor system is one of three systems contributing to postural control (i.e. the control of both

static and dynamic postural equilibrium). The others are the vestibular and visual systems. The cervical muscles, and especially the suboccipital muscles, contain a high percentage of muscle spindles per gram of muscle and these relay sensorimotor information to the central nervous system.^{11, 18, 19} These afferent signals are important for the control of head and neck position and coordination.

Cervicogenic dizziness is caused by altered afferent input from cervical structures in the presence of pain, inflammation, morphological changes or altered muscle activity. Typical symptoms include a vague unsteadiness, light-headedness, nausea and an array of visual symptoms such as blurred vision, the need to concentrate to read, visual fatigue, difficulty judging distances and sensitivity to light.¹⁰⁹ Symptoms are thought to result from conflicting input from the altered cervical afferentation and normal afferent input from the visual and vestibular systems (for reviews, see Clark et al.¹¹⁰ and R  jezon et al.¹¹¹).

It should be noted that these symptoms may also arise from concussion or vestibular dysfunction, or possibly a combination of both. In a blow to the head sufficient to concuss an athlete, the neck and particularly the upper cervical region will also absorb the force and could be injured. Some clinical tests have some differential diagnostic potential^{112, 113} but often it is difficult to differentiate the source unequivocally.

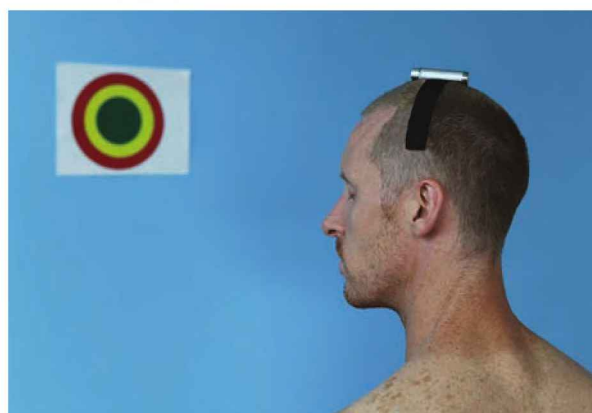
Cervical sensorimotor function may be impaired in any neck pain condition, but is most frequent when the neck pain is a result of trauma or when the athlete complains of such symptoms as light-headedness, unsteadiness or visual symptoms.¹¹⁴⁻¹¹⁷ Three functions are tested: joint position and movement sense, standing balance and oculomotor control.

Joint position sense is measured clinically by assessing the accuracy with which an athlete returns to their natural head posture following movement. Tests of relocation accuracy using rotation in each direction and extension are most common. The athlete sits in a chair positioned 90 cm from a wall to which a pre-calibrated target is attached.¹¹⁸ A laser fastened to a headband is mounted on the athlete's head. The target is positioned on the wall, the centre of which corresponds to the athlete's natural starting position. Following a familiarisation session, the athlete closes his or her eyes, rotates the head as far as possible to the left (at least 45° for a valid test), then returns as closely as possible to the starting position (Fig. 23.10). The test is repeated at least 3-5 times and the mean measurement recorded. Relocation with more than 4.5° of error (the outer red circle and beyond) is outside normal limits.

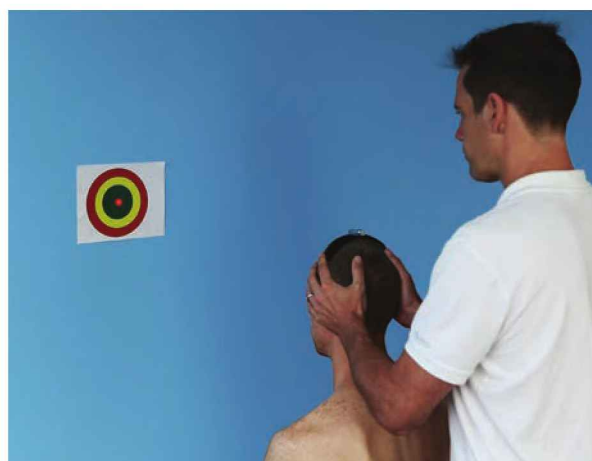
Movement sense may also be tested by having the athlete follow a moving object with the laser pointer.¹¹⁹ Patterns of movement with increasing difficulty are tested, and normative values have been established for each



(a)



(b)



(c)

Figure 23.10 Testing joint reposition sense from rotation (a) The athlete should sit in a supported position with eyes closed throughout the test (b) Each movement direction is tested three times (c) The clinician repositions the head on target between each repetition



Figure 23.11 A test of movement sense. The athlete must trace the shape with the laser as accurately and as fast as possible. Emphasis is placed on accuracy and the patient must keep the laser on the marked lines

scenario.¹²⁰ Alternatively, the athlete can be requested to trace a static target with a laser, with irregular trajectories being assessed (Fig. 23.11).¹²¹

Balance can be measured both statically and dynamically. The cervical sensorimotor system is best challenged with higher-level balance tests than those that may be initially used for a sprained ankle. This is especially so for the athletic population, where balance problems may be overlooked if tests are not challenging enough to detect high-level balance deficits.

Static balance is first assessed by testing the athlete's ability to stand in tandem stance with eyes open for 30 seconds. The test is progressed from eyes open to eyes closed, or from standing on a firm surface to a soft surface (e.g. 10 cm thick dense foam). Further testing progression includes single leg stance or placing the head in a rotated position while in the different stance positions.¹²²

Dynamic balance tests include the conventional Dynamic Gait Index, a 10-metre walk with head turns, or a step test.^{123, 124} These tests could also be progressed to include sport-specific positions and performing multiple tasks. For example, a fencing athlete could be asked to perform an 'advance lunge' while accurately targeting the sword tip. A more dynamic task could be practice of 'balestra', which demands holding a steady end position.

Oculomotor control is tested qualitatively using three tests: gaze stability, the smooth pursuit neck torsion test (SPNTT) and evaluation of rapid eye movements.

Gaze stability is tested by requesting that the athlete fix his or her gaze on an object (e.g. a pen) while comfortably rotating the head to the left and right. The clinician holds a pen about 1 metre away in line with the athlete's nose and observes whether he or she can easily dissociate eye movement from head movement.

The SPNTT evaluates eye follow ability. Impairments in eye follow can be a normal phenomenon or due to neck pain, concussion or a vestibular disorder. The SPNTT was designed to differentiate a probable cervical cause.¹²⁵ In the test, the clinician moves a finger or pen back and forth, crossing the athlete's midline, and assesses eye movement control by watching the athlete's eyes follow the target, noting any saccadic movements (rapid 'catch-up' movements). The test is then repeated with the trunk rotated 45° (the head remaining in neutral position). This targets the neck, but has no effect if the underlying cause is vestibular or due to a concussion. The cervical spine is implicated if eye saccades increase in the turned position compared to the neutral position.¹¹⁶ It is not uncommon for one side only to be compromised. Finally, quickly alternating gaze between two targets (saccadic movements) can also be tested.

Additional tests

Cervical arterial dysfunction

Clinicians who use cervical manual therapy as a treatment modality are very mindful of cervical arterial dysfunction, inclusive of vertebral artery insufficiency and cervical arterial dissections (see Table 23.1). This is reflected in a recently developed international framework that provides safety guidelines for vascular screening prior to hands-on orthopaedic intervention.¹²⁶ Any suspicion of a cervical arterial dissection in progress requires immediate referral to an emergency department.

While any age group could present with symptoms of vertebral artery insufficiency, the risk increases with age due to increased atherosclerosis. Tests are indicated when the athlete complains of symptoms such as dizziness (vertigo), visual disturbances, unsteadiness or other neurological signs. Testing arterial integrity is also recommended as routine before any high-velocity manipulation is performed.

However, clinical provocative tests for the vertebral artery have been shown to have poor validity with low sensitivity.^{127, 128} The sustained rotation and extension tests probably more generally assess the adequacy of collateral flow to the brain rather than the competency of one artery. Yet, they still may have a place to identify the few persons with abnormal vessels in whom collateral compensation is insufficient.¹²⁹ As mentioned previously, of greater concern in the sport context is recognition of CAD (vertebral or carotid) which can be triggered by even minor trauma in the susceptible young to middle-aged athlete.¹²⁹ This history of trauma is a warning signal.

Performance-based outcome measures

Measurements of pain, function and participation levels are important patient-reported outcomes, but equally

important are performance-based outcome measures. These physical measures serve several purposes including testing the immediate effectiveness of a treatment technique, guiding treatment progression, and ensuring that physical impairment associated with the pain and injury has been rectified and the sporting performance skill restored. In addition, showing athletes the results of their individual performance-based outcome measures, in conjunction with objective evidence of improvement over time, can provide powerful incentives for treatment compliance.

A combination of performance-based outcome measures is used for an athlete who has a painful neck condition. An outcome measure should be documented for each impaired system in the athlete's presentation. This could include, for example, a measure of cervical range of motion; a measurement of the upper limb neurodynamic test (range of elbow extension); a measure of motor control (e.g. the cranio-cervical flexion test: stage 1–pressure level achieved with correct movement, and stage 2–pressure level which was achieved and on which training commences); and relocation error in the test of joint position sense. Equally important are performance-based outcome measures of functional skills specific to the athlete's sport. All outcome measures will need to be revisited and upgraded as the athlete progresses in treatment.

MANAGEMENT OF MECHANICAL NECK PAIN

It is usually not possible to make a definitive patho-anatomical diagnosis in most people presenting with a neck pain condition, but this should not compromise conservative management. After red-flag pathologies have been eliminated, the athlete's presenting complaint, the physical examination and any other athlete- or sport-specific issues will direct the treatment approach. In some cases, definitive pathology can be identified, such as a cervical radiculopathy. Alternatively, various syndromes such as a 'stinger', an acute wry neck or a whiplash-associated disorder can provide a label for the neck pain condition. These labels may suggest a certain lesion or underlying structural cause, but the treatment approach should be directed by the examination findings and the athlete's needs. Reference will be made to these syndromes later in this section.

Most athletes fall under the broad diagnosis of mechanical neck pain. The inadequacy of this single classification is that it belies the heterogeneity of neck pain conditions. There are no recipe approaches for mechanical neck pain and each athlete should be regarded as an individual, with management programs designed and tailored to meet their own needs. This section will mention many different treatment approaches. Not all are needed for every athlete. Specific treatment strategies

should be selected based on the current research, the sound clinical reasoning and experience of the clinician, and the unique presentations and goals of the athlete.

Systematic reviews have synthesised and scrutinised the effectiveness of treatment strategies that can be used to reduce pain and disability in the conservative management of neck pain.¹³⁰⁻¹³⁶ Of these various treatments, exercise and manual therapy as single modalities show greatest effect, albeit with modest effect sizes. Used together as a multi-model approach, there is a larger effect.^{132, 133, 136} Other methods of management (e.g. education),¹³⁰ when used as a single-treatment approach, have not been found to be effective. This is hardly surprising considering the nature and extent of all of the impairments or features that may present within biological, psychological and social domains in neck pain conditions, as well as the potential interactions between these domains.

No single modality or method can address such a spectrum. A multimodal approach is essential when neck pain is considered within the biopsychosocial framework. A multimodal approach is summarised in Figure 23.12.

Sport and functional modifications

A comprehensive understanding of the physical requirements of the athlete's sport and training demands, as well as his or her occupational or other recreational activities, is essential to understanding possible provocative factors in order to meet the goals of rehabilitation. If an athlete is performing a task (during sport or work) that is producing or aggravating neck pain, long-term pain relief and prevention of recurrence will not be achieved until this potential 'driver' of the condition is addressed.

For example, an athlete's weight-training regimen is causing neck pain and analysis reveals that when the athlete does a certain lift, his/her head subtly drifts forward. This posture unnecessarily places the upper cervical joints into extension and increases the activity in the neck extensors,¹³⁷ with a net result of added, unnecessary compressive load on the cervical joints. Management must include correction of neck posture to maintain the cervical and cranio-cervical regions in a neutral position during weight training. This may require a temporary reduction in resistance/weight for the lift to allow the neck joints to settle, and rehabilitation of the deep and superficial neck muscles to be effective so that there is adequate muscular support of the cervical region.

As a second example, throwing is aggravating an athlete's neck pain. It is important to consider that there is normal movement of the cervical segments during arm movements.¹⁴ In the presence of neck pain, irregular movements may occur at the cervical segments with throwing and catching actions.¹³⁸ The posture and muscle control of the neck, as well as any deficiency in throwing

Sport and functional modifications

- Review of sport and training techniques
- Appreciate possible provocative factors
- Possible change/correction of technique

Education

- Ensure the athlete understands the nature of their disorder
- Allay any fear or anxiety
- Pain education
- Empower the athlete with knowledge and understanding as a basis for appropriate goal setting

Pain management

- Pharmacotherapies in circumstances of moderate to severe pain or neuropathic pain
- Manual therapy and exercise used for their pain-modulating effects
- Acupuncture, dry needling and trigger point therapy may be useful adjuncts

Manual therapy

- Articular dysfunction is managed using a combination of manual therapy and segmentally directed active exercise
- Techniques may include passive mobilisation, mobilisation with movement and/or manipulation
- Gains from manual therapy must be reinforced with active exercise

Neural tissue mobilisation

- Progress from mobilising the nerve and nerve bed to techniques which tension the nerve and nerve bed
- Manual therapy and active exercise are employed to facilitate the management of sensitised nerve tissue

Training motor function

- Exercise should be prescribed according to the neuromuscular impairments revealed during the clinical assessment
- Exercise are prescribed in two stages. Stage 1 is training to enhance motor control and stage 2 involves resistance training

Training sensorimotor control

- Several physical therapies can improve cervicogenic dizziness
- However, targeted exercises, including proprioception training, balance exercises and oculomotor control exercises, may be superior in reducing measurable impairments

Figure 23.12 Components of a multimodal approach for the management of neck pain disorders. The findings of the assessment will direct precisely the components of management for the individual athlete

technique, should be assessed and addressed in this athlete to eliminate this potential 'motor performance' driver.

A third example involves cycling, which can be provocative for neck pain.³ Despite an ideal bike fit, the neck will inevitably be subjected to non-ideal postures and susceptible to strains in road racing. The athlete is encouraged to reverse the pain-provoking forward-head, upper cervical extension posture as regularly as possible during the ride. Adjusting his postures slightly out of an end-range position may help. In addition, there should be follow-up to ensure that in all other daily activities, the athlete avoids the pain-provoking posture as much as possible.

A final example involves a rugby player with neck pain. Contact sports can be challenging for prevention of neck injury. While rules, equipment and technique may protect

the neck to some extent, it will still be placed under unpredictable forces.^{4, 139-141} A review of the player's rugby tackling techniques or scrum techniques should be a part of the management to lessen the risk, to treat the painful neck condition and to prevent recurrences.

Education

While education alone might not be sufficient for the management of neck pain,¹³⁰ effective multi-modal rehabilitation cannot be delivered without education to ensure the athlete understands the nature of his or her condition. The clinician must demonstrate appropriate interest and empathy, provide assurance and understand the athlete's beliefs about his or her condition as well as his or her coping style, to develop an effective

partnership. Good communication skills are essential and all explanations about the condition should be provided in non-mystical, non-alarmist language.

Pain education is proving to be an important aspect of management. It is also important that the athlete understand what is causing and moderating the condition and how it will be addressed by the management approach. Discuss the evidence for the effectiveness of the proposed management program with the athlete. Education aims to empower the athlete with knowledge and understanding as a basis for appropriate goal setting and stimulates the athlete's participation and compliance with the management program.

Pain management

Pain relief is usually a first priority in people with neck pain and thus most clinical trials have pain as a primary outcome measure, regardless of what treatment is investigated. Pain is an individual, multidimensional sensory and emotional experience, and clinical trials reveal that there are many ways in which the pain experience can be modulated, albeit some more effective than others.

Pharmacotherapies are indicated in circumstances of moderate to severe pain or neuropathic pain. In these circumstances, conservative management or rehabilitation strategies must be limited and not provoke pain, and management can be very challenging for all health professionals. Conservative therapies alone are often not successful for these severe pain states,^{142, 143} and multi-disciplinary management is necessary. Fortunately, most people with neck pain do not have these severe or neuropathic pain states, and conservative management is the typical first-line approach.

Physical modalities including manipulative therapy, exercise and electrophysical agents can modulate pain. Research indicates that a major effect of manual therapy is pain relief through neurophysiological mechanisms.¹⁴⁴ In addition, both low-load and high-load exercises can relieve pain.¹³² Manual therapy and exercise will be discussed in some detail in subsequent sections for, in addition to their effects on pain, they are used for improving motion and enhancing neuromuscular function. Electrophysical agents have little evidence of added benefit,¹⁴⁵ but clinical experience would suggest that they can be quite soothing when part of a multi-modal treatment regimen. There is some evidence for acupuncture, dry needling and trigger-point therapy for immediate and short-term effects on pain.^{135, 146-148} Thus these modalities may also be useful adjuncts to a rehabilitation program.

It is quite normal for some level of anxiety and depression to accompany pain of any origin³³ and this could be especially so if the athlete has had to withdraw from competition. Psychological features can moderate the pain state. Thus appropriate assurance and information

to address such features is an important part of pain management. A decrease in pain parallels a decrease in anxiety and general distress.¹⁴²

Manual therapy

Restriction of neck movement is pathognomonic of painful cervical conditions.⁵⁰ The cervical facet joints and discs are common sources of nociception and movement dysfunction. Articular dysfunction can be managed using a combination of manual therapy (also termed 'manipulative therapy') and segmentally directed active exercise.

Manual therapy techniques generally fall under three categories. The first is passive mobilisation, or rhythmically applied, segmentally directed passive movements. The second is mobilisation with movement, and the third is manipulation, a single high-velocity thrust technique.

There is always concern about cervical manipulation because of the rare possibility of a vertebral or carotid arterial dissection and the associated catastrophic outcomes of stroke or even death.^{126, 129} In response, there has been a recent shift away from the use of purportedly riskier end-of-range cervical rotation techniques.¹⁴⁹

If a clinician has any reservation about the use of cervical manipulation, the evidence suggests that outcomes achieved with cervical mobilisation and manipulation are similar.¹³³ One randomised controlled trial compared outcomes of mobilisation versus manipulation for subacute neck pain. Patients were randomised to manipulation or mobilisation only after the clinician had decided that the patient would benefit from a manipulation.¹⁵⁰ The results showed that even when manipulation was considered to be clinically indicated, but the patient received mobilisation, the outcomes for these patients were the same as those patients who received cervical manipulation. There is no doubt that cervical manipulation can be very effective, but if there is a perceived risk, or the clinician lacks skills to manipulate safely, then the evidence suggests the patient will respond equally well to passive mobilisation.

There are many different theoretical models and clinical approaches to guide specific treatment techniques for the cervical spine, both within and between the different healthcare professionals who use manipulative therapy.¹⁵¹⁻¹⁵³ Evidence continues to emerge that outcomes are similar regardless of the approach used.^{154, 155}

Selection of technique should be based on the examination findings including: the direction(s) of movement restriction; segmental level of pain and dysfunction; pain severity; the nature of resistance to segmental motion (e.g. muscle spasm); and the provisional diagnosis. The athlete must be informed about the approach to be used and consent to the treatment, especially in relation to high-velocity procedures. For the latter, pre-manipulative protocols should be utilised as required by the relevant discipline.¹²⁶

Once an initial technique has been selected, it should be trialled for its effectiveness by assessing its immediate effect on pain and range of motion. Positive within-session changes have been shown to be good predictors of between-session improvements.¹⁵⁶ As indicated previously, mobilisation techniques can be applied passively only, or they can be applied in conjunction with active movement (mobilisation with movement).¹⁵¹⁻¹⁵³

The upper thoracic area should not be neglected as full head rotation is reliant on an approximately 10° contribution from the upper thoracic area.⁷ There is early evidence of improvement in neck pain and range of motion when upper thoracic manipulation is included in the management of neck conditions.^{157, 158}

Manual therapy should be progressed as the patient improves. This can be done in a variety of ways such as increasing the duration of the technique, changing the neck position while the technique is performed using the principles of a combined movement approach,^{159, 160} using a combination of techniques in a single session or, where indicated, adding a high-velocity thrust technique.

Range of motion exercises to accompany manual therapy

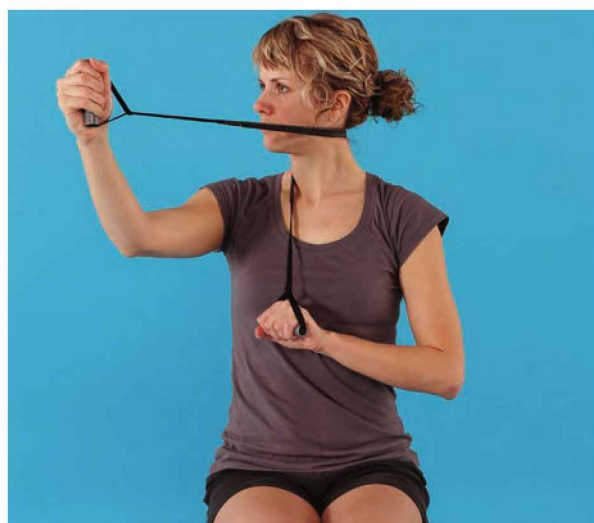
Any gains in range of movement from manual therapy⁵⁰ must be reinforced and maintained with a program of active exercise. Active exercises can be performed in any plane of movement and it is logical to prescribe exercises to target the segment(s) that are hypomobile and that have just been treated. As an example, Figure 23.13 illustrates two self-guided ways that active (a) or assisted active (b) exercise can target (albeit not strictly localise)



(a)



(b-i)



(b-ii)

Figure 23.13 Range of motion exercises (a) Active exercise for lateral flexion. The athlete is taught to localise the segment of interest with his or her fingertip (e.g. if the segment is C2–3, the fingertip is placed on the lamina of C3). The athlete then laterally flexes down to the fingertip to ensure that the segment is moved into lateral flexion (b) The patient uses an assistive strap to localise segmental rotation. For rotation at C1–2, the strap is placed on C1. These techniques are part of the mobilisation with movement regimen¹⁵¹



Figure 23.14 An active exercise which will mobilise rotation throughout the functional cervical spine (C0–1 to T3–4). The athlete imagines shooting a bow and arrow (holding the bow in alternate hands). The athlete must focus on rotation through the upper thoracic region, keeping the mid and lower thoracic regions relatively still

movement to a segmental level. In the acute stage, exercise should be gentle and can be performed three or four times throughout the day as tolerated. It can be progressed in repetitions and intensity as the athlete improves.

Figure 23.14 illustrates an exercise that focuses on mobilising the cervicothoracic region into rotation. It would be an appropriate exercise to prescribe once any acute pain has settled. It is an easy exercise that an athlete can practise throughout the day, which encourages compliance. Alternatively, the combined movements of arm elevation and ipsilateral cervical rotation can also be used to help mobilise the cervicothoracic region.

Neural tissue mobilisation

Treatment of neural tissue mechanosensitivity is a priority when the physical examination confirms that it is contributing to the athlete's neck and arm pain. Some exercises, such as cranio-cervical flexor training, scapular postural correction and certain neck and arm movement exercises, can stress sensitive nerve structures and thereby aggravate pain. Such exercises should not be commenced until the nerve tissue has been addressed.

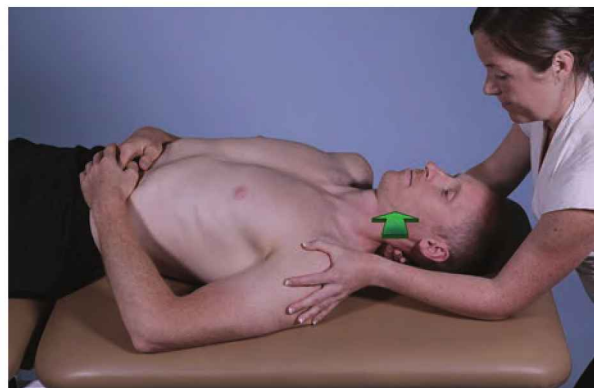


Figure 23.15 Cervical lateral glide technique. If the (L) C6 nerve and brachial plexus are mechanosensitive, the clinician gently maintains the scapular position, grips C6 and glides the head and neck down to C6 to the right. The movement should be gentle and rhythmical. The athlete can contribute to the techniques by gently moving their arm in sequence with the passive technique to enhance the sliding of the nerve and nerve bed

Nerve tissue pain (neuropathic pain) should always be respected, regardless of the presenting pain intensity and the acuteness or chronicity of the condition. Restriction in motion during the upper limb neurodynamic tests is principally due to protective muscle guarding.^{161, 162} The emphasis in contemporary management of neural tissue mechanosensitivity is on mobilising the nerve and nerve bed, with final progression—as the athlete improves—to techniques that tension the nerve and nerve bed.¹⁶³ Both manual therapy and active exercise can be employed in the management of nerve tissue mechanosensitivity.

The cervical lateral glide technique is one efficacious technique (Fig. 23.15).^{164–167} The sliding of the nerve and nerve bed can be enhanced by using combinations of movements to lengthen a nerve bed across one joint while shortening the nerve bed of an adjacent joint. This results in greater nerve excursion with minimal nerve strain and is achieved by the athlete performing the active arm movement while the clinician performs the cervical lateral glide.^{163, 168} For example, in an acute state, the athlete's arm could be positioned by his or her side. As the clinician glides the neck away from this side (lengthens the nerve bed), the patient flexes his or her wrist (shortens the nerve bed). When the clinician returns the neck to neutral (shortens), the patient extends the wrist (lengthens). Progression is achieved by increasing the angle of arm abduction, as well as by flexing or extending the elbow, wrist and hand. A further progression could be to slowly convert the sliding technique to a tensioning technique (i.e. when the neck is glided away, the athlete extends the wrist and hand).

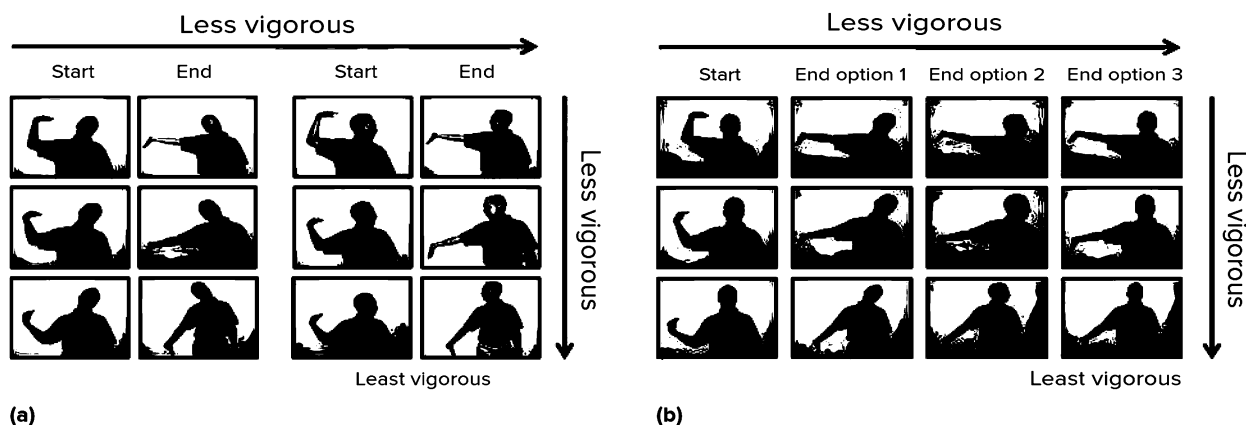


Figure 23.16 (a) Active exercise is used in addition to clinician-delivered technique. The progression for the 'sliding' nerve gliding exercise is illustrated and the level chosen is one that does not provoke the patient's symptoms. The most gentle sliding technique combines neck rotation with elbow and wrist movements while the arm is in 30° abduction. The most vigorous (and the ultimate goal) is to perform the sliding technique in 90° abduction with cervical lateral (top left). The figure illustrates the various transitions from least to most vigorous.¹⁶⁹ (b) The progression for the 'tensioning' nerve gliding exercise. The level chosen is one that does not provoke the patient's symptoms. The most gentle tensioning technique uses elbow and wrist movements with the neck in neutral and the arm in 30° abduction. The most vigorous (and the ultimate goal) is the tensioning exercise in 90° abduction and contralateral lateral flexion (top left). The figure illustrates the various transitions from least to most vigorous.¹⁶⁹

REPRINTED FROM NEE RJ, VICENZINO B, JULL GA, CLELAND JA, COPPIETERS MW. A NOVEL PROTOCOL TO DEVELOP A PREDICTION MODEL THAT IDENTIFIES PATIENTS WITH NERVE-RELATED NECK AND ARM PAIN WHO BENEFIT FROM THE EARLY INTRODUCTION OF NEURAL TISSUE MANAGEMENT. *CONTEMP CLIN TRIALS*. 2011;32(5):760–70 WITH PERMISSION FROM ELSEVIER

This treatment should be supplemented with a home program of active exercise (Fig. 23.16).¹⁶⁹ These exercises are designed to replicate the precise stage of the cervical lateral glide. In relation to dosage, it is prudent to start with low repetitions (e.g. 5 repetitions) two or three times a day to evaluate the athlete's response. Treatment can be progressed by increasing the number of repetitions or altering the arm and neck position, or changing the intention (slide or tension).

Training motor function

A research-informed exercise protocol is used to address the changes identified in the neuromuscular system of the athlete with neck pain (see section on muscle tests above). The neuromuscular and functional changes induced by training are specific to the mode of exercise performed.^{79, 170–173} People with neck pain present with an array of deficits ranging from subtle changes in muscle coordination through to reduced neck strength. Different forms of exercise must be considered and prescribed to address the specific neuromuscular impairments revealed during the clinical assessment. Exercises are prescribed in two stages. Stage 1 is training to enhance motor control. This stage includes exercises to: increase the activation and endurance of the deep cervical postural muscles; improve

coordination between the deep and superficial layers of the neck flexor and extensor muscles and the axioscapular muscles; and train muscle coordination within functional and work activities. Stage 2 involves resistance training and commences once adequate motor control of the region is achieved. As the goal of therapeutic exercise is to restore normal motor control, function and quality of movement, the exercises should be challenging, yet be performed with correct technique and without aggravation of symptoms. Pain can inhibit the motor learning process.¹⁷⁴ For convenience, the rehabilitation of each muscle group is outlined separately, but rehabilitation is concurrent and prescribed based on assessment findings.

Stage 1. Motor control

Neck flexors

Based on assessment findings, any altered coordination between the deep and superficial cervical muscles or decreased endurance capacity of the deep neck flexors must be addressed. Often, athletes must first learn the correct movement of cranio-cervical flexion. They practise a large extension-to-flexion excursion head movement in the supine position, gaining feedback of correct movement by feeling the back of their head sliding up and down the bed. The movement is facilitated with eye movement



Figure 23.17 Training the deep cervical flexors in stage 1. The patient concentrates on performing the correct cranio-cervical flexion movement and palpates to ensure that there is no dominant activity of the superficial flexors. Inner-range contractions are held to improve endurance capacity of these muscles at contraction intensities in line with those required in sitting and standing postures. The pressure biofeedback is used only to teach the patient the holding contraction and to monitor progression of the exercise

and the athlete can palpate his or her sternocleidomastoid and anterior scalene muscles to ensure they are relaxed throughout the entire range of movement (Fig. 23.17). Once the correct movement is attained, endurance of the deep cervical flexors can be trained.

Feedback on performance is necessary so that the athlete knows he or she is performing the task correctly. A pressure sensor can be placed behind the athlete's neck to indicate his or her ability to perform and hold a contraction of the deep cervical flexors (Fig. 23.7). The athlete can practise with the feedback under the clinician's supervision, while learning the sensation of the movement and holding contraction in order to be able to practise at home without the external feedback.¹⁰¹ Training commences at the pressure level that the athlete can control. The athlete practices to achieve 10 repetitions of 10-second holds at that level, twice per day.

Posture

Postural re-education training for spinal and scapular posture commences immediately. The physical examination may have directly linked the athlete's spinal posture to his or her pain state and, importantly for a training effect, the correction of posture facilitates the activity of the deep cervical flexors.^{175, 176} The motor learning process requires multiple repetitions within a day, and postural correction allows the patient to perform multiple repetitions of the muscle contraction during the day with convenience. Furthermore, training the deep cervical flexors in this exercise will help remediate the altered behaviour between

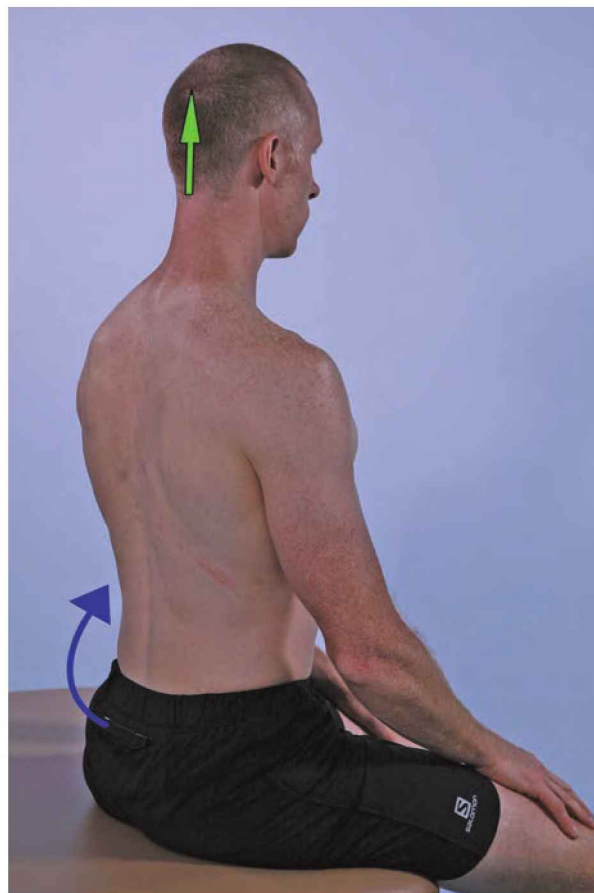


Figure 23.18 Spinal posture correction is facilitated by first correcting the lumbopelvic region into a neutral upright posture. Correct the thoracic posture if necessary, and correct head position by asking the patient to gently lengthen the back of the neck. Once spinal posture has been addressed, the patient will be taught to correct scapular posture as is necessary

the deep and superficial flexors demonstrated in the CCFT.¹⁷⁷ An upright posture is facilitated by first moving the lumbopelvic region into a neutral position (Fig. 23.18). This will usually correct thoracic posture, although a slight correction of excessive thoracic flexion or extension may be necessary. As a final correction, the athlete should gently lengthen the back of the neck, which will not only facilitate the longus colli further,¹⁷⁶ but will correct the cervical and cranio-cervical position.

The athlete is advised to perform this posture correction exercise ideally 2-3 times per hour. They should hold the position for 10 seconds. The exercise can be incorporated into daily activities while sitting, standing and walking. Repetition is the key and the athlete must identify or develop 'cues' as reminders for the exercise.

Formal training of cranio-cervical flexion continues until the patient can perform 10 repetitions of 10-second holds at each of the five pressure increments on the CCFT from the baseline of 20 mmHg to the maximum of 30 mmHg. Once this goal for the deep cervical flexors is achieved, the program is progressed and the interaction between the deep and superficial flexor muscles is trained during movement. One of the movements to train is control of neck extension. This movement is initiated with cranio-cervical extension and return to the upright position from extension is initiated by cranio-cervical flexion. Concomitantly as required, control of neck position in any sport-specific position is trained.

Neck extensors

Stage 1 training for the neck extensors commences in inclined sitting, prone on elbows or, most commonly, in a quadruped position, depending on the athlete's ability to control scapular and trunk posture. The athlete can train the movements of cranio-cervical extension and rotation and cervical extension (C2-7) (Fig. 23.8a and b). Instructing the athlete to read an imaginary book (eyes focused down between hands) while curling the head and neck backwards maintains the cranio-cervical region in a neutral position, facilitating the desired cervical extension (Fig. 23.8c). Manual resistance can be applied by the clinician in extension over the vertebral arch to emphasise the activation of the deep extensor muscles while familiarising the athlete with the exercise.¹⁷⁸ Athletes should train until they can perform three sets of 10 repetitions through a full excursion of movement without fatigue.

Axioscapular muscles

Stage 1 training focuses initially on improving the activation and endurance capacity of the trapezius and serratus anterior muscles as well as re-educating the axioscapular muscles functionally to provide adequate scapular stability in posture and during arm movements.

In this motor relearning phase, the lower portion of trapezius can initially be targeted with the athlete in a side-lying (gravity-eliminated) position with the arm supported on pillows at approximately 140° of elevation. The athlete practises drawing the scapula diagonally across the thoracic wall and holding the position for 10 seconds (10 repetitions). Serratus anterior is formally trained with a scapular protraction exercise which can be progressed as indicated from a side-lying position, to standing against a wall, to a prone position on elbows, to a four-point kneeling position. The exercises in the latter two positions can be combined with neck extension exercises. The scapular muscles are also trained concurrently in the postural correction exercise and the athlete learns to hold

the scapulae in a neutral position on the chest wall in sitting and standing postures.

Once the athlete can control 'ideal' static scapular posture, scapular control is re-educated for functions requiring less than 30° of arm elevation (control of the scapula in a neutral position), during arm movement throughout range and, importantly, in sport- or occupation-specific arm postures and movements.

Stage 2. Resistance training

Stage 2 addresses any deficits in muscle strength and demands greater levels of endurance. Neck flexor strength and endurance are trained at progressive contraction intensities by increasing the gravitational load using a head lift exercise.

Initially, the athlete sits on a chair close to a wall with his or her head resting on the wall in a neutral position. Cranio-cervical flexion is performed by sliding the back of the head up the wall, and then the head weight is lifted just off the wall (Fig. 23.19). Cranio-cervical control is essential in this exercise. If the patient loses cranio-cervical flexion when the head is lifted, this means that the exercise load is too high. The load is increased by progressively moving the chair further from the wall and ultimately moving to crook-lying and supine positions. The exercise can be further progressed in supine from having support of two pillows to no pillows to performing the exercise with weighted sandbags on the forehead.

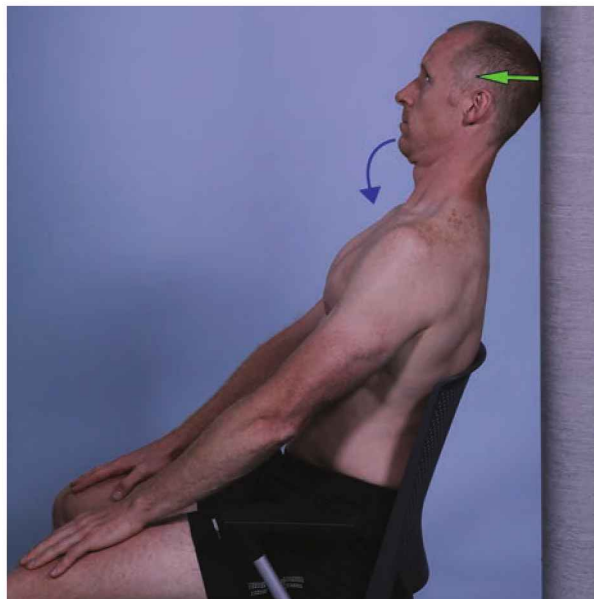


Figure 23.19 Head lifts from the wall. Note that the patient must control the cranio-cervical flexion position in this initial strengthening exercise in stage 2 training

For the cervical extensors, the athlete can perform repeated cervical extension or sustained neck extension holds in a prone or quadruped position. Progressive weights (commencing with 0.5 kg) can be attached to headgear (e.g. a bike helmet) if neck-specific resistance equipment is not available. For the axioscapular muscles, free-weight or machine-based exercises can be performed. Strengthening exercises should commence with light loads concentrating on spinal and scapular control with both concentric and eccentric arm loading. Strengthening exercises should be task-specific with a focus on upper limb movements that are relevant to the athlete's sport. Load is progressed, but the emphasis is always placed on control of scapular and spinal posture, and indeed all body segments. Resistance training for each muscle group is advanced as appropriate to the individual athlete's requirement (e.g. neck strength requirements for pistol shooting versus wrestling or rugby).

When the aim is to enhance strength, high loads should be prescribed for a low number of repetitions, whereas a large number of repetitions at low intensity will enhance endurance. A key principle in resistance training is that of overload.¹⁷⁹ This principle states that a greater-than-normal stress or load on the body is required to induce training adaptation. As performance improves with training, the intensity or volume of exercise must be increased to continue to put adequate demand on the athlete to stimulate muscle development.

Training sensorimotor control

There are several treatment options that can improve symptoms of dizziness, lightheadedness and unsteadiness of cervical origin. These include specific and targeted exercises as described below which directly address the impairments,¹⁸⁰⁻¹⁸³ as well as treatments such as manual therapy, exercise and acupuncture.^{154, 183} There is some evidence that the targeted exercises may be best at making a measurable improvement in these impairments.¹⁸⁰

Joint relocation sense can be trained with eyes open, progressing to eyes closed with use of a laser and target for feedback. Similarly, a target can be traced with a laser to train movement sense progressing from simple to complex patterns (Fig. 23.20). Following a moving target is another treatment strategy, and various computerised applications such as training with virtual reality are being developed to train cervical movement sense.^{120, 182, 184} The task difficulty can be progressed in the training of both position and movement sense by changing the size or stability of the base of support (i.e. sitting to standing to tandem stance to one-foot stance).

When training balance in an athlete with neck pain, it is usually necessary to commence training at a fairly high level; for example, in tandem stance. This can be progressed



Figure 23.20 Movement sense is trained with the athlete tracing simple to complex patterns with a laser light mounted on the head. The exercise can also be progressed by advancing the position in which the exercise is performed (e.g. from sitting to standing to one-foot standing)

to one-leg standing and performed on hard, then soft and then unstable surfaces. Ensure that training is progressed to the functional demands of the athlete's sport.

Training eye movement control could include any of the following strategies: gaze stability, eye follow and eye follow of rapid movements; it should be based on the impairments detected in the physical examination. The tests described above can be converted to training

exercises, with the clinician ensuring that the athlete can dissociate head and eye movement in all exercises. The exercises are usually commenced at a speed over which the athlete has control and they are progressed by increasing exercise speed or performing the exercises in more challenging postures (e.g. sitting to standing to tandem stance) as done when training joint relocation sense. Athletes with neck pain often need more advanced exercises, and combining training of two aspects of the cervical sensorimotor system (e.g. eye follow and balance) will produce a greater challenge to the cervical sensorimotor system.

Maintenance program

Specific self-management and maintenance programs should be included within the athlete's training regimen. A self-management program developed during treatment should incorporate motor relearning, resistance and sensorimotor exercises as indicated. As previously mentioned, repetition and adequate dosage are vital components to the success of any training program. Neck pain is a recurrent condition. With the possibility of re-injury during sport or the potentially continuing influence of arthrogenous inhibition,¹⁸⁵ it is necessary for the athlete to continue to perform a simple maintenance program after formal management has ceased. For long-term compliance, this should consist of two or three key exercises aimed at reducing the probability of recurrent neck pain episodes.

NECK PAIN CONDITIONS

The majority of neck pain conditions are grouped under the broad term 'mechanical neck pain' and then qualified by a description of duration as being acute, subacute, persistent or recurrent acute episodes. Management is based on examination findings, as discussed throughout this chapter. Certain neck pain conditions are further characterised in relation to a specific clinical presentation, injury mechanism, identifiable pain mechanism or pathology. Management will again be based on examination findings, but some additional notes are relevant from a diagnostic or treatment perspective.

Cervicogenic headache

Cervicogenic headache is classified as a secondary headache (Chapter 21): it is secondary to a cervical musculoskeletal condition.¹⁸⁶ Pain sources are usually located in one or more of the upper three cervical segments and headache results from physiological convergence between cervical and trigeminal afferents in the trigeminocervical nucleus.¹⁸⁷ There is symptomatic overlap between migraine without aura headaches, tension-type headaches and cervicogenic headaches, and thus differential diagnosis is critical for proper management.

Research has sought to determine which physical tests characterise cervicogenic headache to differentiate it from migraine and tension-type headaches. In cervicogenic headaches, the athlete should present with a combination of cervical movement dysfunction, muscle function impairments and upper cervical joint dysfunction.^{49, 55, 188} The clinical findings in the articular and neuromuscular systems are consistent with mechanical neck pain, and any reported unsteadiness or lightheadedness is consistent with sensorimotor disturbances. Management generally follows that described earlier in this chapter for mechanical neck pain, but care should be taken to ensure that treatments are non-pain provocative. This is because sensitisation of the trigeminocervical nucleus appears to be a factor in the development of cervicogenic headache.⁵⁷ If treatments are painful, the added peripheral nociception is likely to produce or worsen a headache.

Acute wry neck

Acute wry neck is characterised by an acute onset of neck pain with protective deformity (contralateral lateral flexion position) and limitation of movement. It typically occurs after a sudden, quick movement or on waking. There may have been an unguarded movement or prolonged abnormal posture prior to the onset of pain. It is hypothesised that the zygapophysial joint or disc may be the source of symptoms. When of zygapophysial origin, there may be entrapment or extrapment of the meniscus.¹⁸⁹ If the disc is suspected to be the source of symptoms, treatment is more conservative (i.e. gentle mobilisation; manipulation is not recommended). The differentiation between a zygapophysial joint or disc origin of symptoms is principally made collectively from historical features, presenting symptoms and manual detection of a pain source either over the zygapophysial joint or centrally through the spinous process, which is interpreted as a discal origin.

Although relatively rare, the clinician should be alert for atlanto-axial rotary fixation, a fixed rotary subluxation of C1 on C2.^{190, 191} It is principally observed in young children and may occur spontaneously or can be secondary to congenital anomalies of the upper cervical spine and arthritides (e.g. juvenile onset rheumatoid arthritis). It may be associated with trauma and there may be instability. Medical and radiological diagnosis is required.

Cervical nerve injury Cervical radiculopathy

Cervical radiculopathy usually presents as neck and arm pain or, occasionally, arm pain only, and may be accompanied by sensory disturbances and weakness in the upper limb. This neuropathic pain can often be

intense and athletes require appropriate pharmacological pain management. The cervical nerve root can be compromised in the intervertebral canal by a disc lesion, or by zygapophysial or uncovertebral osteophytes. Inflammation also plays a significant role. Sensitisation of nerve tissue will be present. Surgery may be indicated in some instances, but conservative treatments should always be trialled in the first instance.¹⁹²

Brachial plexus injuries ('stingers/burners')

Brachial plexus injuries occur in contact sports such as rugby union and rugby league, and can vary from severe injuries with avulsion or rupture¹⁹²³ to less catastrophic traction injuries, often referred to as 'stingers' or 'burners' due to their characteristic symptoms.¹⁹⁴

Assessment of an acute stinger involves first ruling out more serious injury such as spinal cord or brain involvement, fracture or vascular injury.¹⁹⁵ The athlete should be removed from the field if there are any such indications. In a stinger or burner, the athlete experiences transient neuropraxia characterised by upper extremity burning-type pain that may also be accompanied by paraesthesia and/or weakness. Tensile or compressive forces are the most likely underlying mechanism of this neuropraxia. Downward displacement of the shoulder with concomitant contralateral side flexion of the neck can cause a traction injury to either the cervical nerve root (most likely C5 or C6) or the brachial plexus. Alternatively, compression injuries can be caused by combined cervical extension and ipsilateral side flexion or a direct blow to the brachial plexus.¹⁹⁴

Symptoms are usually transient, resolving over a varying period of time from minutes to days, but persistent neurological dysfunction and recurrent stingers may occur. A gentle exercise program that does not provoke typical

symptoms may be started early in treatment, with care to avoid aggravating positions. As in all cases of nerve damage, there will be an element of the recovery that is time- and rest-dependent, and treatment cannot speed this process.

Acceleration/deceleration injury ('whiplash')

Acute acceleration/deceleration injury to the cervical spine is an injury commonly caused by a motor vehicle collision and it is commonly termed a whiplash-associated disorder (WAD). Theoretically, such an injury could also occur in sports when the cervical spine is subject to a sudden acceleration force. However, direct extrapolation from injuries sustained in motor vehicle collisions to a sporting injury may be unwise, and more comparative research is required. Management of most athletes with WAD will be similar to mechanical neck pain, but there is a small proportion of those with WAD who may not respond well to any treatments (medical, physical or psychological) and who generally have a poor prognosis.^{25, 196-198}

CONCLUSION

Mechanical neck pain conditions can be troublesome for any athlete, but most can be managed well with conservative therapies. Nevertheless, neck pain is characteristically recurrent and its lifetime impact should not be underestimated. Management must aim to first address the presenting condition and resolve the acute episode of neck pain. The challenge is the second stage of treatment that aims to prevent or limit the recurrence of neck pain to improve future quality of life and sports performance. This broader aspect of effective secondary prevention should be a focus of both clinical practice and research.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

